ABSTRACT

SHAH, PARIN PRATIK. Aerodynamic Modeling, Experimental Validation, and Dynamic Analysis of a Spherical Sailing Omnidirectional Rover (SSAILOR). (Under the direction of Dr.Andre Mazzoleni).

The Spherical Sailing Omnidirectional Rover (SSailOR) is a lightweight, rolling spherical rover designed for navigation in remote and harsh environments, including planetary exploration, Arctic monitoring, and long-duration surveillance missions. The SSailOR consists of a dual-sail arrangement and a traction hoop (similar to a sailboat's keel), which is used not only for generating forward thrust but also for providing heading control and the ability to maneuver relative to the prevailing wind direction. This work aims to get a broader understanding of SSailOR's interaction and performance under varying conditions that mimic real-world operating environments. A dynamic model of the SSailOR is formulated using Kane's method, accounting for all acting forces and variable parameters, including sail angle of attack, SSailOR yaw angle, and wind effects. An aerodynamic model for the sails using a vortex lattice method (Weissinger method) is developed to capture the coupled aerodynamic effects generated by the two sails when operating together. The configuration was evaluated across multiple heading angles and through a systematically varied combination of main sail and jib sail angle of attack. These variations were selected to replicate the range of orientations SSailOR may encounter during directional maneuvering and upwind sailing modes. This model was verified using XFLR5, a widely used aerodynamic analysis software that also employs vortex-lattice and panel methods. Comparing results between the Weissinger implementation and XFLR5 provides an important measure of confidence in the model and helps identify conditions where linear assumptions begin to break down. Wind tunnel tests are conducted on the SSailOR under varying wind speeds and yaw angles to characterize the aerodynamic behavior of the entire rover, not limited to the sails. The findings revealed dominant sources of aerodynamic drag, leading to design changes that reduced drag and enabled the idea of improved upwind motion through a higher lift-to-drag ratio. The results from the wind-tunnel tests are validated using computational fluid dynamics (CFD) simulations performed in ANSYS Fluent. The correlation between CFD predictions and wind tunnel measurements served as a key indicator of the reliability of both approaches. Collectively, these results validate the proposed modeling and experimental approach and demonstrate SSailOR's potential for reliable, controlled upwind mobility in real-world environments.

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Aerodynamic Modeling, Experimental Validation, and Dynamic Analysis of a Spherical Sailing Omnidirectional Rover (SSAILOR)

by
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DEDICATION

To my family for their endless love, support, and encouragement.

BIOGRAPHY

Parin Shah was born in Mumbai, Maharashtra, India, on October 7, 2002. He earned his Bachelor's degree in Mechatronics Engineering from Mukesh Patel School of Technology Management and Engineering in May 2024. He enrolled at the North Carolina State University to pursue a Master's degree in Aerospace Engineering. He then joined the Engineering Mechanics & Space Systems Lab, directed by Dr. Andre Mazzoleni, and started working on the Spherical Sailing Omnidirectional Rover (SSAILOR) project, where he focused on aerodynamic modeling, dynamic modeling, experimental testing, and design optimization. Outside of work, Parin enjoys watching cricket and traveling.

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AUTHORSHIP STATEMENT

CHAPTER

1

INTRODUCTION

As we move towards a new era of planetary and space exploration, with rapid technological advancement and novel scientific discovery, the boundaries of innovation continue to expand. From the Earth's icy polar regions to the surfaces of distant extraterrestrial bodies, the demand for persistent, high-quality data collection in extreme and remote environments has never been greater. However, such environments, including the Arctic and extraterrestrial terrains, present significant challenges for conventional exploration methods, where human access is limited, costly, or altogether impractical. To address these challenges, there is a growing need for autonomous, unmanned robotic systems capable of operating reliably under harsh and uncertain conditions. Mobile robotic platforms offer a promising solution by enabling sustained data collection, environmental monitoring, and surveillance without direct human intervention.

Some sources of inspiration include solar-powered rovers like Solero (Michaud et al. (2002)) and Cool Robot (Ray et al. (2007)), and tumbleweed rovers, which can traverse large distances by harnessing wind energy. While solar-powered rovers have proven effective for long-duration missions, their performance is often constrained by terrain complexity and energy availability. The rovers presented in Wilson et al. (2008), Hartl and Mazzoleni (2008), Hartl and Mazzoleni (2010), Mazzoleni and Hartl (2012) rely on drag-based propulsion for motion. Although these designs are highly energy efficient, drag-driven propulsion inherently limits controllability and

prevents upwind travel. A detailed comparison of these concepts and the associated trade-offs is presented in Varanwal et al. (2026). Motivated by this literature review, a new spherical rover concept has been proposed and is currently under development.

The Spherical Sailing Omnidirectional Rover (SSailOR) is a first-of-a-kind wind-powered autonomous rover capable of net upwind motion. As shown in Fig 1.1, this is a spherical rover inspired by a sailboat. Its traction hoop serves as the centerboard, enabling movement on the given terrain. Long-duration mobility is enabled by wind propulsion via onboard sails. These sails are controllable for desired paths, including upwind motion. Potential applications for this concept are wide, including planetary exploration, arctic observation, and military surveillance. This rover is to have a design that generates lift with minimal energy consumption. This is a unique concept, which makes it complex and requires a well-designed rover.

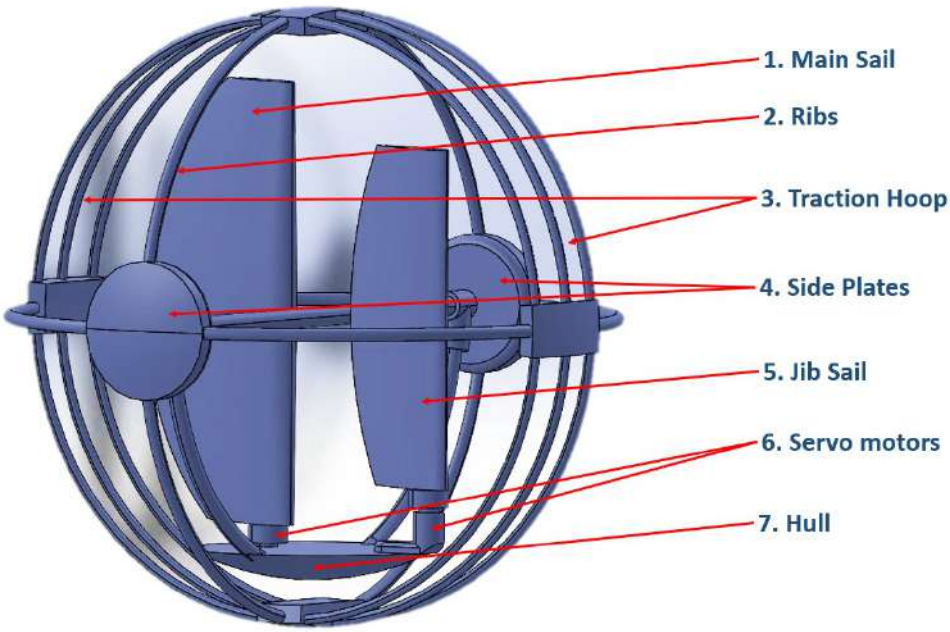


Figure 1.1: Schematic Diagram of SSailOR

1.1 Background

Traditionally, most sailboats consist of a mainsail and a jib sail, which work together in tandem to produce a slot effect that enhances airflow and increases lift compared to a single sail configuration. As seen in Fig 1.2, the jib accelerates the flow over the mainsail by reducing

flow separation and improving pressure recovery, resulting in higher aerodynamic efficiency. Additionally, sailboats employ a keel that counteracts lateral forces generated by the sails, providing directional stability and enabling efficient upwind motion. These characteristics of a dual-sail configuration help achieve controlled propulsion and maneuverability in adverse wind conditions.

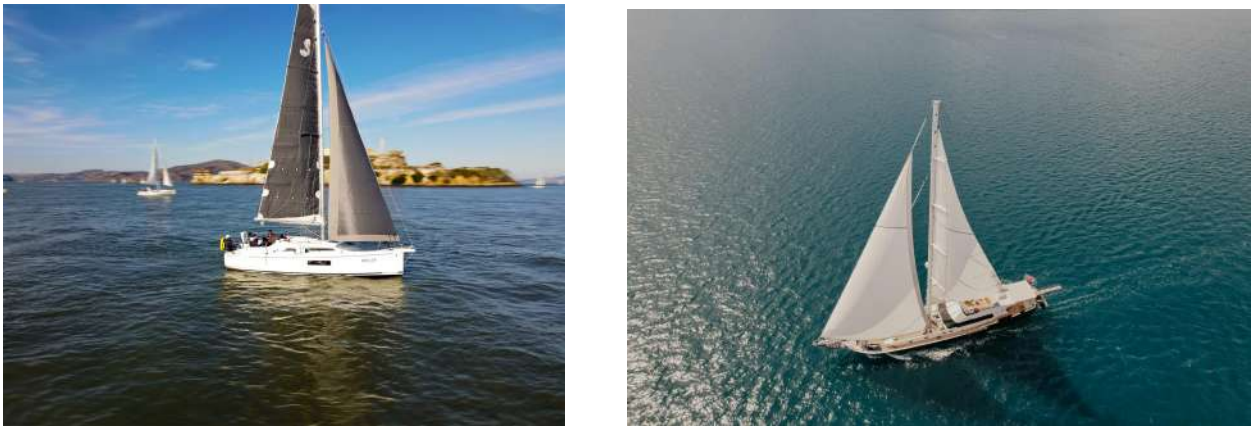


Figure 1.2: Mordern Sailboats

The Spherical Sailing Omnidirectional Rover (SSailOR) follows a similar aerodynamic principle but adapts it to a ground-based planetary rover system. Instead of utilizing hydrodynamic resistance from a keel, SSailOR employs a traction hoop to resist lateral motion while converting aerodynamic forces into controlled rolling motion of the spherical body. The rover integrates a dual-sail configuration mounted on an internal hull structure enclosed within a spherical shell. Like conventional sailboats, the interaction between the jib and mainsail creates a slot effect that enhances lift generation and improves aerodynamic efficiency, enabling propulsion even in off-wind and upwind conditions. By leveraging aerodynamic lift and drag forces generated through the dual-sail configuration, SSailOR demonstrates the feasibility of wind-driven planetary exploration systems capable of sustained and energy-efficient traversal in environments where conventional wheeled rovers may face operational limitations. Table 1.1 shows the comparison of other wind-powered rovers with the SSailOR presented in Kosak et al. (2026).

Table 1.1: Comparison of wind-based rovers with SSailOR

Rover	Self-Righting	Upwind-Capable
Land Sailer	NO	YES
Tumbleweed Rover	YES	NO
SSailOR	YES	YES

1.2 Motivation and Objective

The performance of SSailOR under varying wind conditions is evaluated using a dynamic model. This model enables the prediction of rover behavior in a controlled simulated environment that closely replicates real-world operating scenarios. The dynamic model accounts for the dominant forces acting on the rover, with primary emphasis on the aerodynamic forces generated by the dual-sail configuration. These aerodynamic forces are computed using a numerical aerodynamic model, allowing evaluation across a wide range of mainsail and jib sail angles of attack. Additionally, the model captures variations in rover heading relative to the apparent wind direction, enabling analysis of propulsion efficiency and achievable upwind motion.

While numerical models provide valuable insight into system behavior, experimental validation is essential to ensure model performance. To achieve this, wind-tunnel experiments were conducted using a 40% scaled SSailOR prototype. The scaled testing approach enables controlled measurement of aerodynamic forces, lift-to-drag characteristics, and slot-effect interactions between the sails while maintaining manageable manufacturing cost and testing complexity. These experiments provide critical experimental data to validate aerodynamic predictions and identify potential flow phenomena that may not be fully captured through analytical or numerical methods alone.

Complementary Computational Fluid Dynamics (CFD) simulations were performed to further validate and interpret wind-tunnel measurements. CFD analysis enables detailed visualization of flow structures, pressure distributions, and interaction effects between the dual sails that are otherwise difficult to measure experimentally. By correlating CFD results with wind-tunnel data, the study establishes a strong aerodynamic validation framework. This combined experimental and computational approach strengthens the reliability of the aerodynamic model and provides a robust foundation for guiding future full-scale SSailOR development and operational deployment.

1.3 Thesis Structure

This thesis is organized into six chapters, each addressing a key aspect of the aerodynamic characterization and dynamic modeling of the SSailOR.

Chapter 1 introduces the motivation and background of wind-driven systems and discusses the aerodynamic principles behind dual-sail configurations. It also outlines the research objectives and overall methodology adopted in this study.

Chapter 2 presents the experimental testing of SSailOR through wind tunnel testing. Two separate wind-tunnel tests were conducted using a scaled SsailOR prototype. The first test was performed to find out the aerodynamic performance of an initial design of the SSailOR and identify its limitations. Based on observations from the initial testing, design optimizations were implemented to improve aerodynamic efficiency. A second wind tunnel test was then conducted to evaluate the effectiveness of these modifications. This chapter also includes a detailed comparison between the results obtained from both testing campaigns, highlighting performance improvements and design insights.

Chapter 3 focuses on Computational Fluid Dynamics (CFD) analysis performed to complement and validate experimental testing results. This chapter describes the CFD setup, including geometry preparation, meshing strategies, boundary conditions, and solver selection. The CFD results are analyzed to obtain detailed flow-field characteristics and aerodynamic force predictions. A comparative analysis between CFD and wind tunnel results is presented to assess model accuracy and aerodynamic performance trends.

Chapter 4 presents the development of a numerical aerodynamic model for the SSailOR's sail configuration using the Vortex Lattice Method (VLM), specifically employing the Weissinger method. This chapter describes the modeling assumptions, discretization strategy, and aerodynamic force calculations. To validate the numerical model, results obtained from the VLM are compared with a separate analysis performed using XFLR5, providing an additional level of verification for the aerodynamic analysis.

Chapter 5 introduces the dynamic modeling of SSailOR using Kane's method. This chapter describes the formulation of the rover's multibody dynamic model, including the selection of generalized coordinates, derivation of kinematic relationships, and development of equations of motion. The resulting dynamic model is implemented and simulated to evaluate rover motion under varying conditions.

Chapter 6 summarizes the key findings of the thesis and presents conclusions drawn from experimental, computational, and analytical investigations. This chapter also outlines potential areas for future work, including model refinement, implementing control strategies, and full-scale rover development.

CHAPTER

2

WIND TUNNEL TESTING AND DESIGN OPTIMIZATION

As mentioned in 1.3, two separate wind tunnel tests were conducted on different design iterations of SSailOR. The first wind tunnel test, referred to as Test 1, was performed on the initial rover design to establish baseline aerodynamic performance. The data collected from this campaign was systematically analyzed to identify aerodynamic limitations, flow interaction effects, and opportunities for design improvement. Based on these findings, several design modifications were implemented to enhance aerodynamic efficiency and overall performance. The updated configuration was subsequently evaluated through a second wind tunnel test, referred to as Test 2, which was conducted to assess the effectiveness of the implemented design changes and quantify performance improvements relative to the initial design.

2.1 Test 1 Setup

The Test 1 setup, as shown in Figure 3.1a, consisted of a test rig incorporating upper and lower mounting structures, sensors for measuring rover orientation, SSailOR rotational speed (RPM), and wind speed within the test section. The rig also included motors for rover and sail rotation, along with aerodynamic fairings designed to minimize wake formation generated by

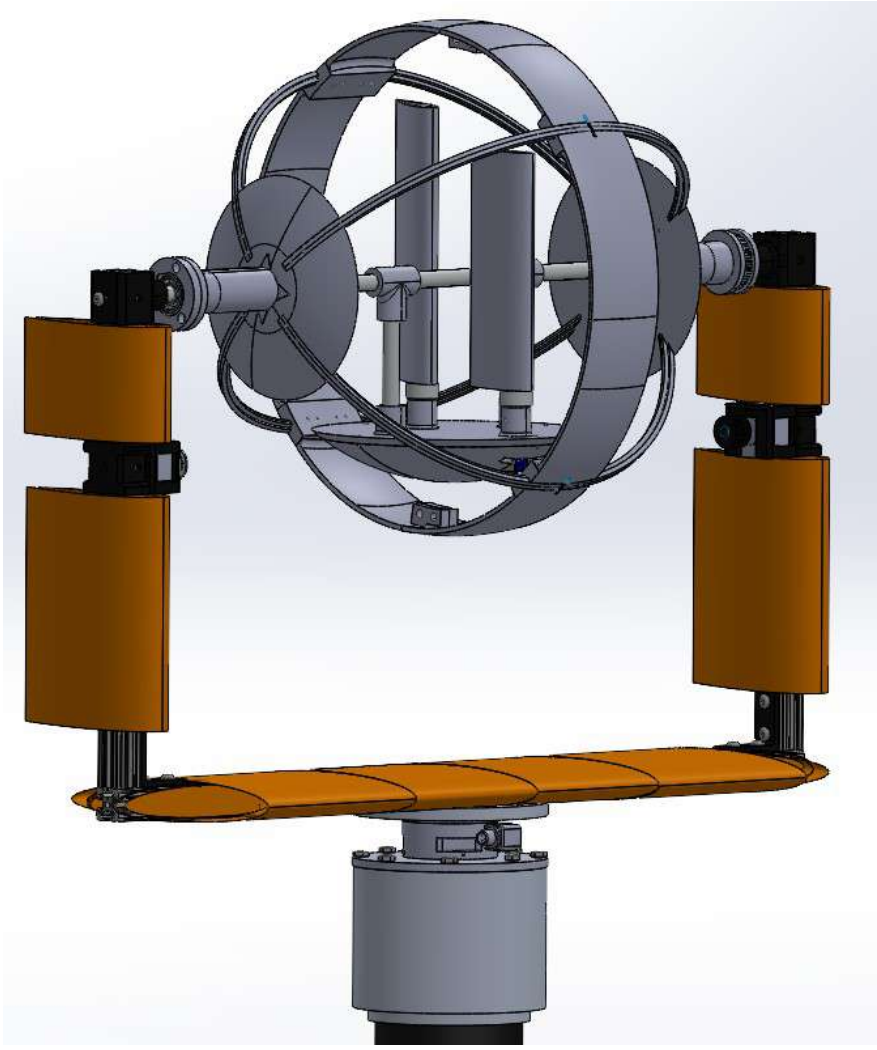


Figure 2.1: Test Mount For SSailOR (CAD Model)

the supporting structure. Structural symmetry was maintained across the upper assembly using vertical fairings and a dummy motor to ensure uniform mass distribution and aerodynamic balance. The entire setup was designed to fit within the wind tunnel test section, which has dimensions of 32 × 45 × 46 inches. Considering wind tunnel blockage constraints, a half-scale SSailOR prototype with a diameter of 40 cm was selected to ensure accurate aerodynamic measurements while maintaining acceptable blockage ratios. System Parameters in more detail are shown in Table 2.1.

To minimize flow disturbances caused by the support hardware, an 80 mm clearance was maintained between the rover and the supporting structures. The structural frame of the test rig was primarily constructed using 80-20 T-slot aluminum extrusions, which, while providing modularity and structural flexibility, introduce bluff-body geometries that can generate

significant wake and turbulent flow structures. Such disturbed or dirty airflow can alter the velocity profile and reduce flow uniformity reaching the SSailOR, thereby affecting aerodynamic measurements. To mitigate these effects, vertical and horizontal aerodynamic fairings were integrated into the support structure. Vertical fairings with symmetric airfoil cross-sections and horizontally oriented elliptical fairings were implemented to streamline the support frame and reduce pressure drag and wake formation. Minimum wall and floor clearances were maintained in accordance with standard wind tunnel testing practices to limit boundary layer interference and wall effects. The NACA 0030 symmetric airfoil and elliptical fairing geometries were selected based on structural integration requirements, expected flow velocities, and wake reduction performance.

Table 2.1: System Parameters

Parameters	**Values**
Rover Diameter	400 mm
Side Plate Height	19.3 mm
Side Plate Diameter	174.4 mm
Main Sail Chord	100 mm
Main Sail Span	215 mm
Jib Sail Chord	70 mm
Jib Sail Span	185 mm
Hull Diameter	215 mm
Ribs T section width	8 mm

2.1.1 Mechanical Setup

Mobility

All rover components were fabricated using FDM 3D printing. The assemblies utilize standardized T-slotted rails, nut-and-bolt fasteners, and M2 screws to maintain modularity. The hoop structure is assembled from multiple arc segments joined using interlocking 3D-printed sleeves. The spherical body is supported by a central aluminum rod equipped with four bearings to enable low-friction rotation. Drive torque is applied through a stepper motor and timing belt connected to a driven gear, with the selected gear ratio being 2:1.

Frame

The main structural frame includes T-slot rails, a central aluminum rod, and 3D-printed connectors to ensure mechanical isolation of the rover from external disturbances. The vertical fairings rotate with the yaw axis and are fixed in position using mechanical collars. The horizontal fairings are directly attached to the lower frame. Structural symmetry was maintained by installing a second stepper motor on the opposing side, which remained unpowered during testing.

Sensor Placement

The load cell is mounted beneath the test rig, between the test frame and the lower motor mount, ensuring accurate force and torque measurements. The sphere's rotation is verified using a Hall-effect magnetic sensor mounted opposite the drive assembly. This sensor reads a rotating magnet attached to a cylindrical extension connected to the rover frame. An IMU sensor is placed at the lower portion of the test section, below the load cell. A wind sensor is mounted on the upper wall of the wind tunnel section, positioned upstream of the test rig.

2.1.2 Electronics Setup

The electronics test rig was designed to enable synchronized motor control, sensing, and data acquisition for controlled wind-tunnel experimentation. A micro-stepping driver controlled the stepper motor's speed and position through external commands, enabling precise rotation of the spherical rover body. The rotational speed of the sphere was measured using a Hall-effect sensor that generated one pulse per revolution. Free-stream wind velocity was measured using an upstream wind sensor calibrated specifically for the wind tunnel test section. Yaw orientation of the rover was monitored using two Inertial Measurement Units (IMUs) positioned to capture system orientation.

Table 2.2: Communication Protocols For Sensors

Sensor	**Communication**	**Connection**
IMUs	I^2C	Microcontroller
Hall Effect Sensor	Digital INT	Microcontroller
Wind Sensor	UART	Microcontroller
Load Cell	AI Signal Output	NI PXIe AI channel
Servo Motor	PWM	NI PXIe CO channel
Stepper Motor	PWN	NI PXIe CO channel
Microcontroller	Serial	Labview via USB

The IMUs communicated with the microcontroller via the I^2C communication protocol, allowing synchronized acquisition of angular position and rate data. The Hall-effect sensor output was received through a digital interrupt channel on the microcontroller to ensure accurate RPM measurement. The wind sensor transmitted data via UART, which was read directly and processed by the microcontroller. Closed-loop control of the jib sail was implemented using real-time torque feedback from the six-axis load cell. Complete list of sensors used and their connection is mentioned in Table 2.2. A servo motor actuated the jib sail angle to maintain near-zero net torque about the vertical axis under varying aerodynamic loading conditions. The six-axis load cell simultaneously recorded the forces (Fx, Fy, Fz) and moments (Mx, My, Mz), thereby providing a complete characterization of the aerodynamic loads.

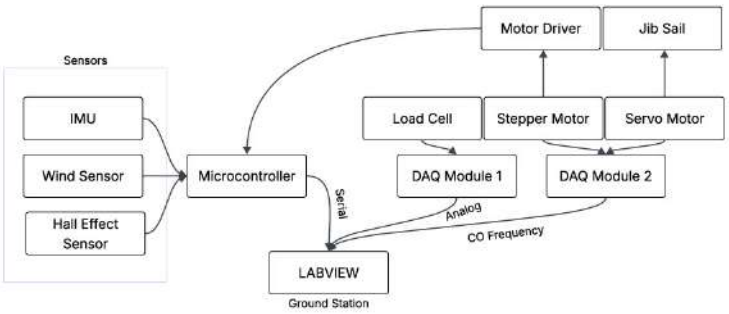


Figure 2.2: Block Diagram representing SSailOR's Electronics and DAQ

Sensor and actuator signals were integrated through two primary acquisition pathways, as shown in Fig 2.2. A microcontroller served as the central sensing interface, collecting IMU data via I^2C, Hall-effect sensor data via digital input interrupts, and wind sensor measurements via serial communication. The microcontroller performed time stamping and preliminary signal conditioning before transmitting synchronized data streams to the ground station using serial communication. High-precision analog and control signal acquisition was performed using a National Instruments PXIe-based DAQ system. The load cell analog outputs were routed

to analog input (AI) channels of the PXIe DAQ module for force and moment measurements. Motor control signals for both the servo motor and stepper motor driver were generated using pulse-width modulation (PWM) via the counter output (CO) channel of the DAQ system, enabling precise actuator control.

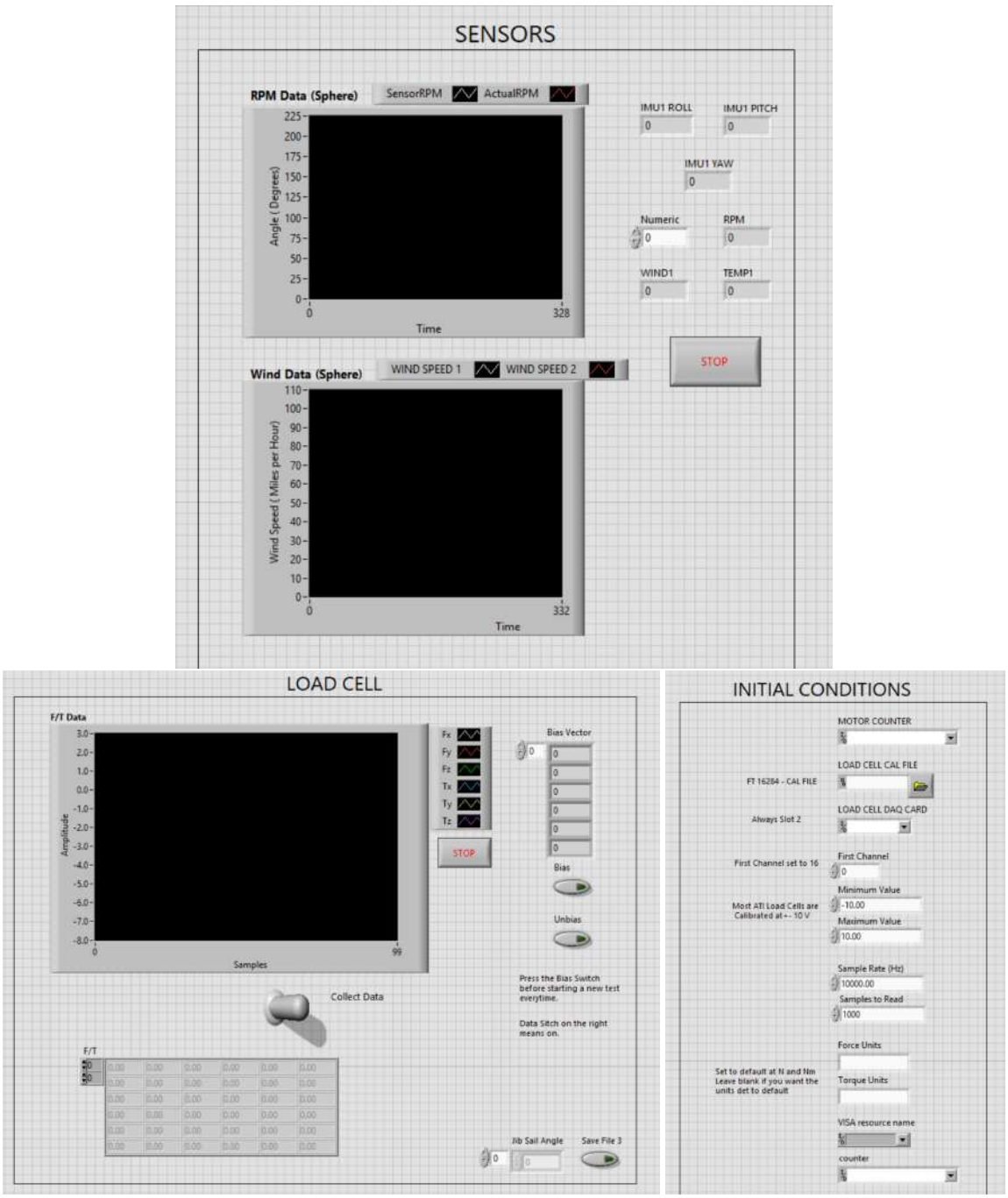


Figure 2.3: Labview Front Panel

LabVIEW functioned as the primary ground station interface, providing real-time visualization of sensor data, actuator states, and overall system monitoring, as shown in Figure 2.3. Data logging was implemented within LabVIEW using a user-controlled acquisition trigger. Upon activation of the recording switch, LabVIEW automatically initiated data capture and generated structured Excel files for data storage. All sensor and actuator channels were recorded at a sampling rate of 100 Hz, ensuring high-resolution capture of transient and unsteady aerodynamic phenomena.

2.2 Test 1 Matrix

2.2.1 Baseline Testing

Full-system tests were conducted under a range of operating conditions to measure aerodynamic forces and moments as functions of wind speed, yaw angle, sail configuration, and rover rotational speed (RPM). These experiments provided a comprehensive characterization of SSailOR's aerodynamic performance under representative wind-driven operating scenarios. In addition to free-stream testing, a ground-effect study was performed by positioning a flat steel plate beneath the rover model, shown in Figure 2.4b. The plate was supported by four NACA 0025 airfoil struts to minimize structural wake interference while maintaining geometric stability. This configuration enabled systematic evaluation of how proximity to the ground surface influences aerodynamic force and moment measurements. Together, these test parameters form the foundation for assessing SSailOR's aerodynamic behavior and performance under realistic environmental conditions. The primary operating conditions evaluated during baseline testing are summarized in Table 2.3.

Table 2.3: Main Test Parameters

Parameters	**Values**
RPM	0, 60, 80
PSF	1, 1.5
Yaw	10°, 20°, 30°, 40°
Main Sail AoA	-8°, -4°,0°, 4°, 8°

2.2.2 Sub-System Testing

To isolate the aerodynamic contributions of individual components, tests were conducted under three configurations: (i) structural frame only, (ii) hoop and ribs assembly, and (iii) sails with inner hub. Upon analysis of the final results, the measured structural forces were subtracted from the total recorded forces to isolate the net aerodynamic forces acting solely on the SSailOR configuration. The test conditions used for subsystem-level evaluation are provided in Table 2.4.

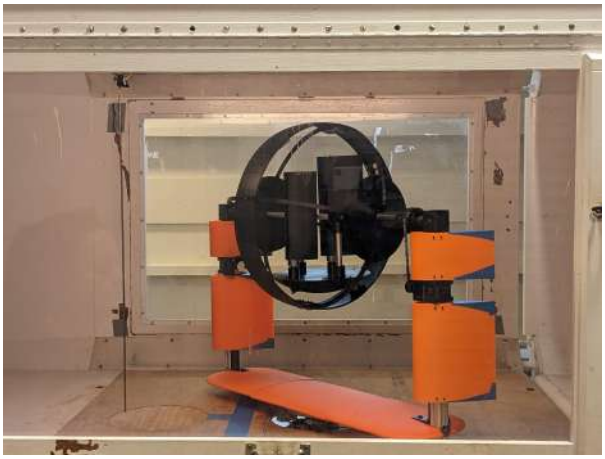
Table 2.4: Sub-Systems Test Parameters

Case	**RPM**	**PSF**	**Yaw** (°)	**Main Sail AoA** (°)
Structure Only	N/A	1.0, 1.5	10, 20, 30, 40	N/A
Ground Effect Test	60, 80	1.0, 1.5	10	4, 8
Hoops + Ribs	0, 80	1.5	20, 30, 40	N/A
Sails + Hub	N/A	1.5	20, 30, 40	0, 4, 8, 12

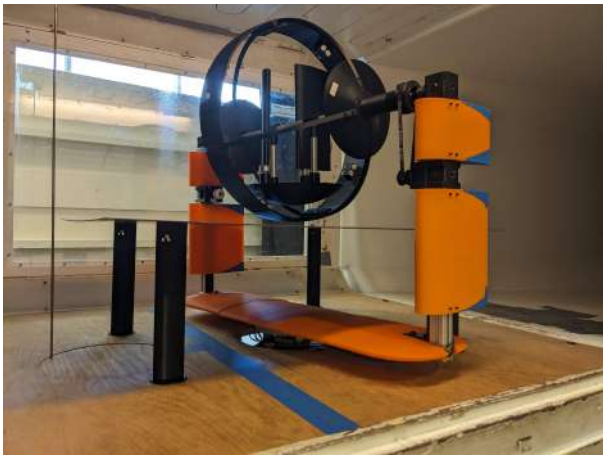
2.3 Test 1 Results

For Design 1, wind tunnel test results at 1 PSF are shown in Fig. 2.5 for different yaw positions, while the corresponding results at 1.5 PSF are shown in Fig. 2.6. Due to the solid traction hoop geometry in Design 1, the drag forces were significantly high. Also, Design 1 had smaller sail sizes (Table 2.5), which resulted in lower lift forces. Served as baseline design, Design 1 has given meaningful and important insights which were used during Design 2 development. To understand the coupled aerodynamic behavior of Design 1 sails, separate tests were conducted to get the lift and drag forces on sail + hull configuration, mentioned in Kosak et al. (2026) Varanwal et al. (2025).

Also, a limited set of tests was conducted with a ground plane installed beneath the rover to assess sensitivity to ground proximity effects Varanwal et al. (2025). Comparison of aerodynamic forces obtained with and without the ground plane showed a relatively small difference in force magnitudes, while the overall trends remained unchanged. This indicates that, under the tested conditions, ground plane effects are secondary to the aerodynamic forces generated by the rover and that the absence of a ground plane in most Design 1 tests was acceptable for capturing the dominant aerodynamic behavior.

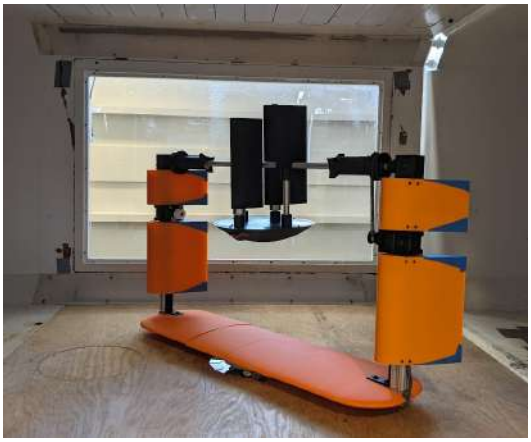


(a) SSailOR inside Wind Tunnel – Test 1



(b) Ground Effect Test

(c) Only Structure Test



(d) Only Sails Test

Figure 2.4: Wind tunnel experimental configurations for SSailOR during Test 1.

2.4 Design Optimization

Following the results obtained from Test 1, several aerodynamic and structural modifications were implemented to improve overall performance. The initial configuration exhibited unfavorable lift-to-drag characteristics and excessive parasitic drag from non-lifting structural components. These observations motivated a redesign aimed at increasing aerodynamic efficiency while reducing unnecessary flow obstruction.

Design 2 introduced targeted geometric and architectural improvements to enhance the lift-to-drag ratio. The primary modifications included: 1) Implementation of a segmented three-ring traction hoop architecture to reduce projected frontal area and wake interference. 2) Increase in total sail area to improve lift generation capability. 3) Reduction in hull frontal area to decrease form drag. 4) Replacement of T-shaped ribs with circular cross-sections to

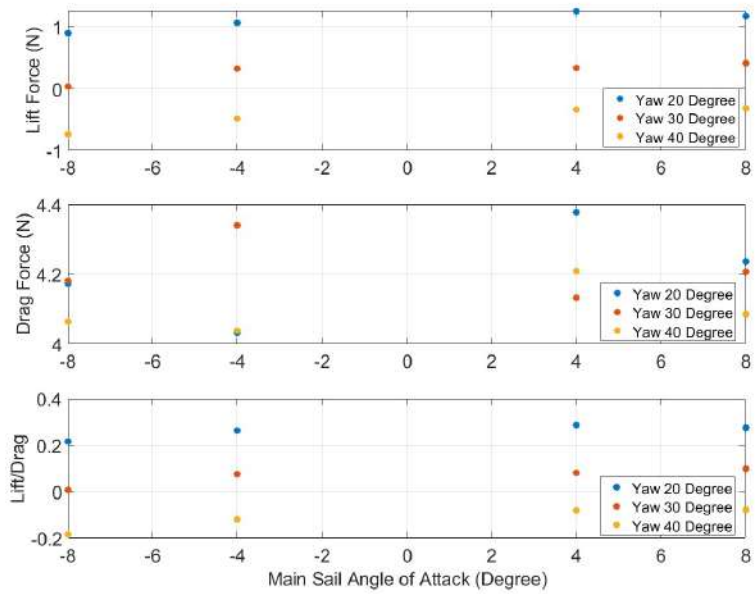


Figure 2.5: Design 1 Wind Tunnel Test Results at 1 PSF

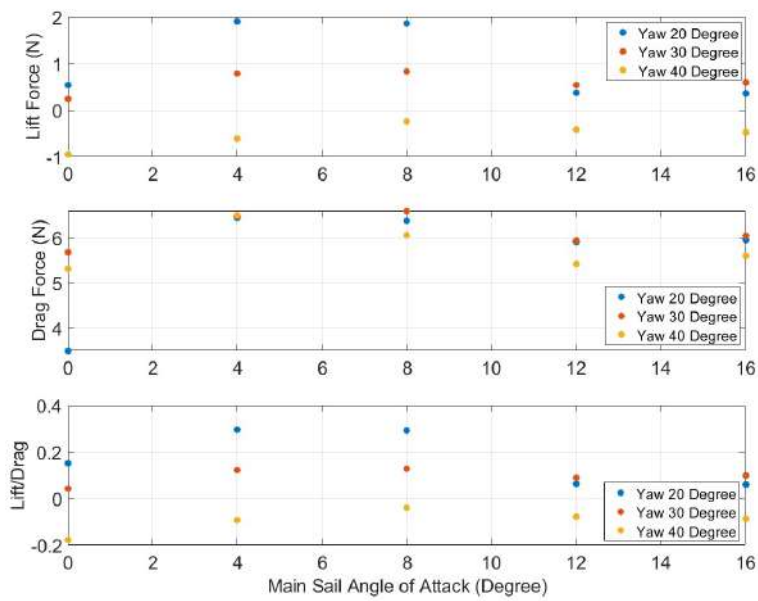
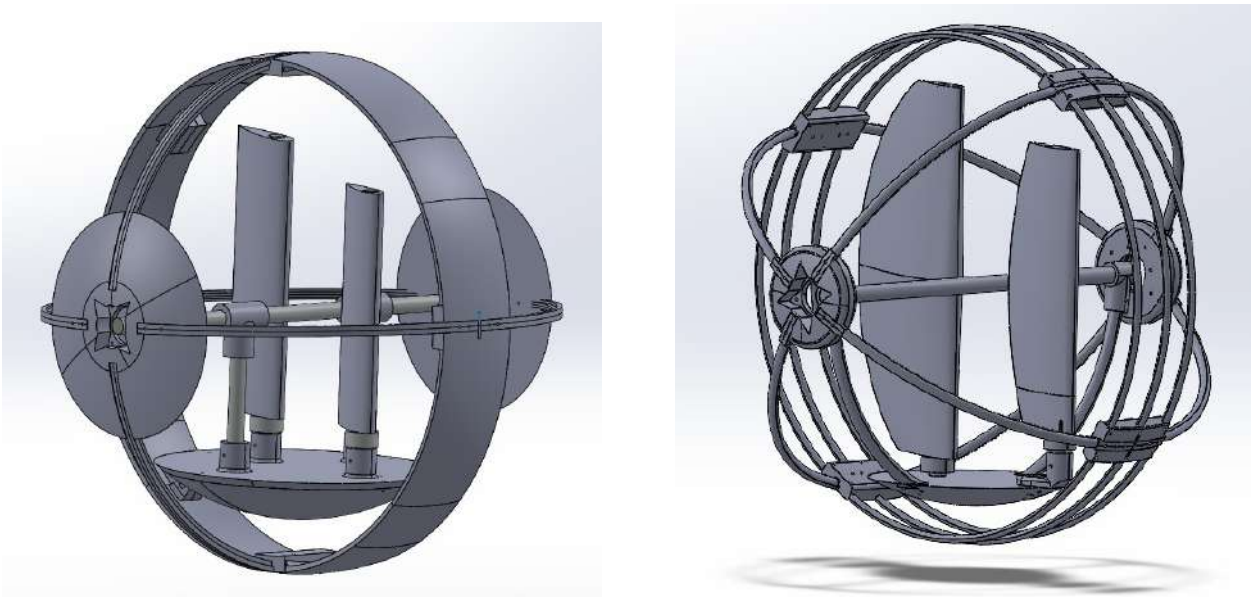


Figure 2.6: Design 1 Wind Tunnel Test Results at 1.5 PSF

minimize bluff-body drag.

A comparison of the two configurations is summarized in Table 2.5 and is also illustrated in Figures 2.7a and 2.7b. The changes in sail area, aspect ratio, structural components, and frontal area form the basis for the experimental performance comparison presented in the



(a) Design 1 CAD (Baseline Design)

(b) Design 2 CAD (Improved Design)

Figure 2.7: Comparison of Design 1 and optimized Design 2

subsequent sections.

Accurate aerodynamic characterization of SSailOR is essential for high-fidelity dynamic modeling and closed-loop control development. Reliable lift and drag coefficients are required not only for the main and jib sails but also for non-lifting components such as the traction hoop, ribs, side plates, and hull. Due to the geometric complexity of the rover and the coupled aerodynamic interactions between components, simplified analytical estimates are insufficient for implementation-level modeling. Therefore, the optimized configuration was subjected to a second wind tunnel campaign (Test 2) to experimentally quantify the impact of these design improvements.

2.5 Test 2 Setup

The experimental setup for Test 2 was identical to the configuration described in Section 2.1, except for the updated SSailOR geometry corresponding to Design 2. The same wind tunnel facility, mounting structure, load cell arrangement, sensors, and data acquisition system were utilized to ensure consistency and enable direct comparison with Test 1 results.

The optimized rover configuration, incorporating the segmented three-ring traction hoop, increased sail areas, reduced hull frontal area, and circular rib cross-sections, was mounted onto the existing modular test rig, as shown in Figure 2.8. Care was taken to maintain identi-

Table 2.5: Comparison of Design 1 and Design 2 Geometric and Structural Features

Parameter	**Design 1**	**Design 2**	**Unit**
Main-sail airfoil	NACA0020	NACA0020	–
Main-sail area	0.0215	0.0392	m^2
Main-sail *AR*	2.15	2.10	–
Jib-sail airfoil	NACA0025	NACA0020	–
Jib-sail area	0.0129	0.0201	m^2
Jib-sail *AR*	2.64	2.41	–
Traction hoop	Solid	3-Ring design	–
Hull frontal area	0.0061	0.0022	m^2
Rib cross-section	T-shape	Circular	–

cal boundary conditions, clearance distances, and instrumentation placement to eliminate experimental bias between the two test campaigns.

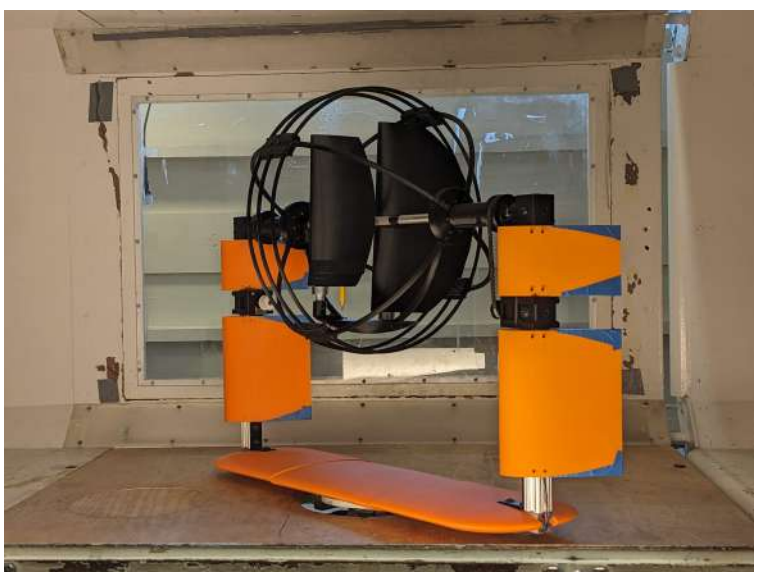


Figure 2.8: SSailOR with Design 2 in Wind Tunnel

2.6 Test 2 Matrix

The Test 2 parameter matrix closely followed the operating conditions defined in Test 1 to enable direct performance comparison between the two rover configurations. Dynamic pressure levels (1 PSF and 1.5 PSF), yaw angles, and RPM settings were maintained consistent with the baseline campaign.

However, negative main sail angles of attack were excluded from Test 2. Results from Test 1 indicated that negative sail angles did not produce meaningful propulsion-oriented lift behavior and therefore were not relevant to the optimized configuration's operational envelope. Consequently, Test 2 focused exclusively on non-negative sail angles of attack to evaluate performance within realistic operating conditions.

Additionally, subsystem-level testing conducted during Test 1 (structure-only, hoop + ribs, sails + hub configurations) was not repeated. The objective of Test 2 was to assess the full-system aerodynamic performance of the optimized design rather than isolate individual component contributions. This streamlined test matrix allowed efficient evaluation of design improvements while maintaining consistency with baseline conditions.

Table 2.6: Test 2 Main Test Parameters

Parameters	**Values**
RPM	80
PSF	1, 1.5
Yaw	20°, 30°, 40°
Main Sail AoA	0°, 4°, 8°, 12°, 16°

Table 2.6 summarizes the operating conditions evaluated during Test 2. The matrix was designed to maintain comparability with Test 1 while focusing exclusively on physically meaningful sail orientations.

2.7 Test 2 Results

The lift and drag force measurements for Design 2 are presented in Fig. 2.9 and Fig. 2.10. Compared to the baseline configuration (Test 1), the optimized design shows clear improvements in aerodynamic performance across all tested operating conditions.

For both dynamic pressures (1 PSF and 1.5 PSF), Design 2 consistently generates higher lift forces over the full range of main-sail angles of attack and yaw angles. The increase in

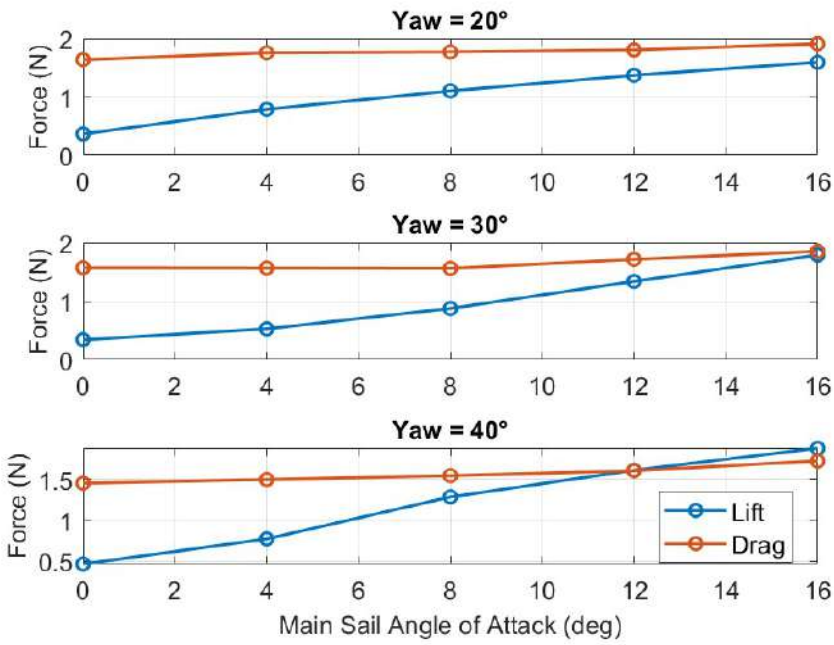


Figure 2.9: Wind Tunnel Test 2 Results at 1 PSF

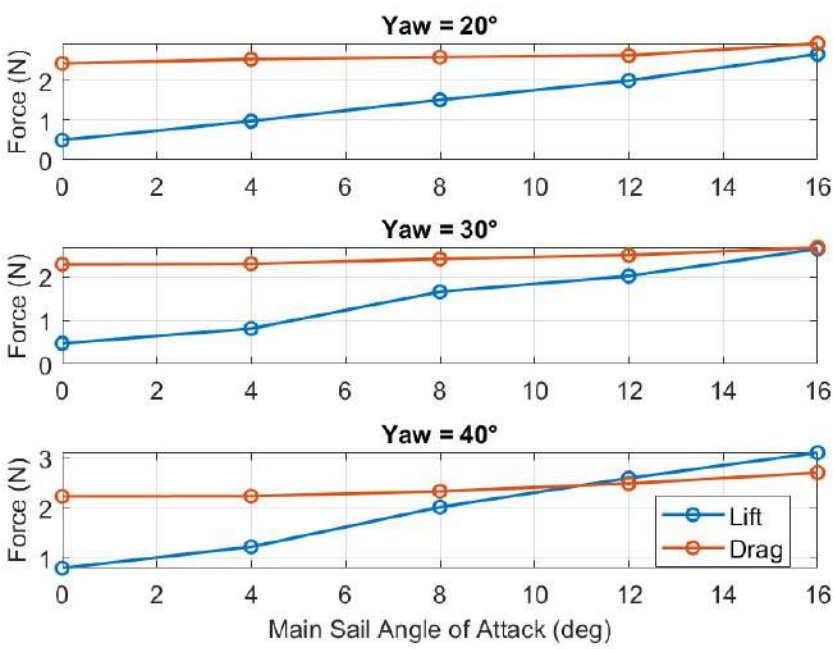


Figure 2.10: Wind Tunnel Test 2 Results at 1.5 PSF

lift is particularly pronounced at larger yaw angles, where aerodynamic interaction between the main sail and jib plays a dominant role. This improvement is primarily attributed to the increased sail area and reduced flow obstruction introduced in the optimized configuration.

In addition to enhanced lift generation, Design 2 demonstrates a measurable reduction in drag force relative to the baseline design. The drag reduction is largely due to the segmented three-ring traction hoop architecture, which replaces the solid hoop used in Design 1. This modification reduces frontal area and minimizes bluff-body wake formation, thereby decreasing parasitic drag contributions from non-lifting structural components.

The force trends further indicate that the flow field interacting with the main and jib sails is less disturbed in Design 2. The open traction hoop configuration allows cleaner incoming flow to reach the sail surfaces, improving aerodynamic efficiency. This effect is most evident at higher yaw angles, where coupled sail interactions are more sensitive to upstream flow disturbances.

Overall, the combined increase in lift and reduction in drag leads to a higher lift-to-drag ratio for Design 2 across the tested operating envelope. Since the lift-to-drag ratio is a critical parameter for sustained upwind motion and energy-efficient propulsion, these results validate the effectiveness of the design optimization strategy. Consequently, Design 2 was selected for subsequent aerodynamic coefficient identification and dynamic modeling.

2.8 Conclusion

When lift and drag force results are compared for Design 1 and Design 2, a clear qualitative difference between the two designs emerges. Design 2 consistently produces higher lift forces while keeping the drag lower than that of Design 1. These clearly indicate that the design change from Design 1 to Design 2 clearly improved aerodynamic performance.

Beyond the observed force improvements, Design 2 incorporates a segmented three-ring traction hoop design that more closely represents the intended final rover configuration. In contrast, the solid traction hoop used in Design 1 was primarily intended for early-stage testing and does not reflect the geometry planned for continued development. The segmented design reduces flow obstruction and provides a more realistic representation of the aerodynamic interactions expected in the final prototype.

Based on both the improved aerodynamic force behavior and the greater design relevance of the traction hoop configuration, subsequent aerodynamic coefficient identification and modeling efforts are focused exclusively on Design 2. Design 1 therefore serves as a reference configuration for comparison, while Design 2 forms the basis for the aerodynamic models developed in the remainder of this work.

CHAPTER

3

CFD ANALYSIS

3.1 Introduction

For CFD study and comparison, testing was conducted to span a wide range of operational conditions. These tests considered variation of the rover's yaw position ψ, sail angle of attack α_M and α_J, wind speed V_∞ (in terms of dynamic pressure Q), and cage rotational speed RPM. The main sail angle of attacks was extended beyond its stall angle to figure out how the rest of the rover may affect it. CFD simulations were made in parallel with the wind tunnel test 2. Each test method followed the same test matrix, created from the variables in table 3.1.

Table 3.1: Test Variables With Varying Jib

Parameter	**Values Tested**	**Units**
Test configurations per design	30	–
Rover heading (yaw)	20, 30, 40	degree
Main sail angle of attack	0 – 16	degree
Jib sail angle of attack	0 – 18	degree
Freestream wind speed	9.178, 11.241	m/s
Reynolds number	$1.5 \times 10^5 - 2.5 \times 10^5$	–

3.2 CFD Study

3.2.1 Methodology

Computational Fluid Dynamics (CFD) simulations were performed using ANSYS Fluent to investigate the aerodynamic behavior of SSailOR. ANSYS Fluent solves the Reynolds-Averaged Navier–Stokes (RANS) equations to model viscous fluid flow and is widely used in aerospace applications for external aerodynamic analysis. Extensive literature on this can be found in Estrada et al. (2021); Zawawi et al. (2018); Suresh Babu et al. (2024).

For this study, the Shear Stress Transport (SST) k–ω turbulence model was employed. The SST k–ω model combines the near-wall accuracy of the k–ω formulation with the free-stream robustness of the k–ε model, making it well-suited for flows involving adverse pressure gradients and possible separation. Given the dual-sail configuration of SSailOR and the likelihood of localized flow separation around the sails and structural components, the SST model provides improved predictive capability compared to standard turbulence models.

The simulations were configured to closely replicate the wind tunnel setup. The computational domain matched the wind tunnel test section dimensions, and the rover geometry corresponded to the same scaled model used in experiments. Two primary boundary regions were defined: one representing the wind tunnel enclosure and the other representing the rover surfaces. The motion of the traction hoop was neglected in this static analysis to focus on steady-state aerodynamic forces.

Two inlet velocity conditions corresponding to the wind speeds used in the wind tunnel experiments were prescribed. Unlike the experimental setup, where airflow conditions were characterized using dynamic pressure (PSF), the CFD simulations required specification of inlet velocity, temperature, and air density. A pressure outlet boundary condition was applied downstream, and no-slip wall conditions were enforced on all rover and tunnel surfaces.

This steady-state CFD analysis provides aerodynamic force predictions for comparison with wind tunnel measurements and numerical modeling results. A more comprehensive transient CFD study incorporating rotational effects and unsteady flow behavior is currently in progress.

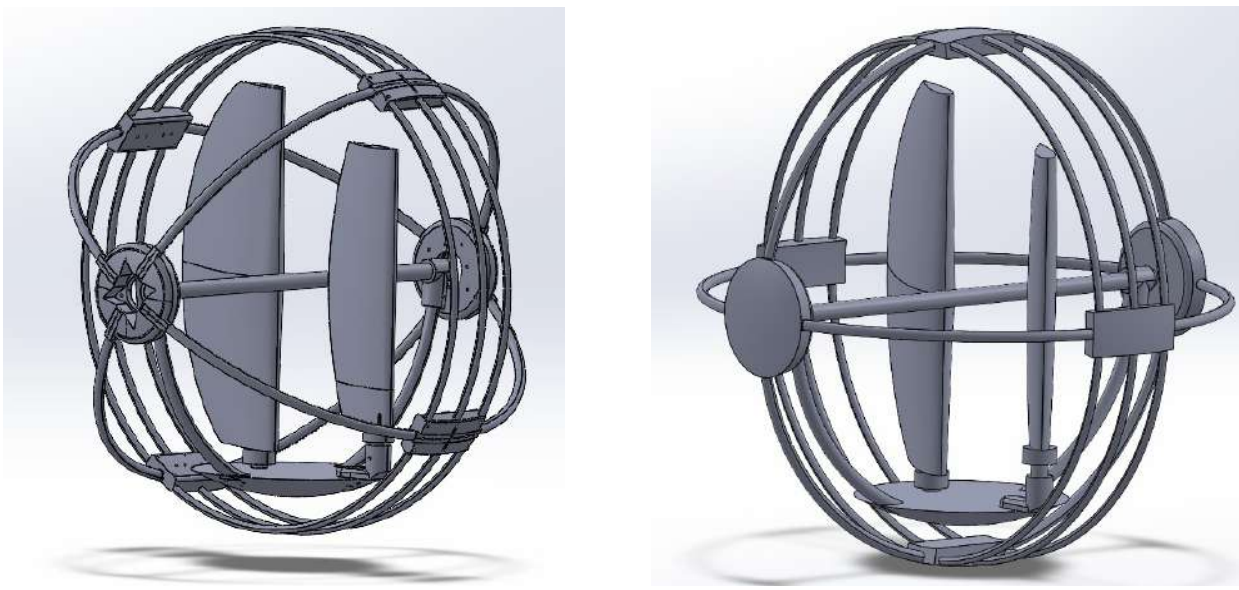
3.2.2 CFD CAD Model

For the CFD simulations, the computational geometry closely represented the physical SSailOR prototype used in wind tunnel testing, as shown in Fig. 3.1a. However, several modifications were necessary to ensure compatibility with the CFD solver and to improve mesh quality.

Non-manifold geometries and small geometric inconsistencies were removed to prevent meshing errors. Screw holes and minor fastener features were filled, as these small-scale details

do not significantly influence global aerodynamic behavior but substantially increase mesh complexity. The side plates were modified to provide smoother transitions to the side ribs, reducing sharp discontinuities that could generate unnecessary numerical artifacts. Additionally, traction hoop connectors were simplified to eliminate overextended or problematic geometric features that could degrade mesh quality.

The resulting CFD-ready geometry, after simplification and cleanup, is shown in Fig. 3.1b. These modifications preserved the overall aerodynamic characteristics of the rover while ensuring robust mesh generation and stable solver convergence.



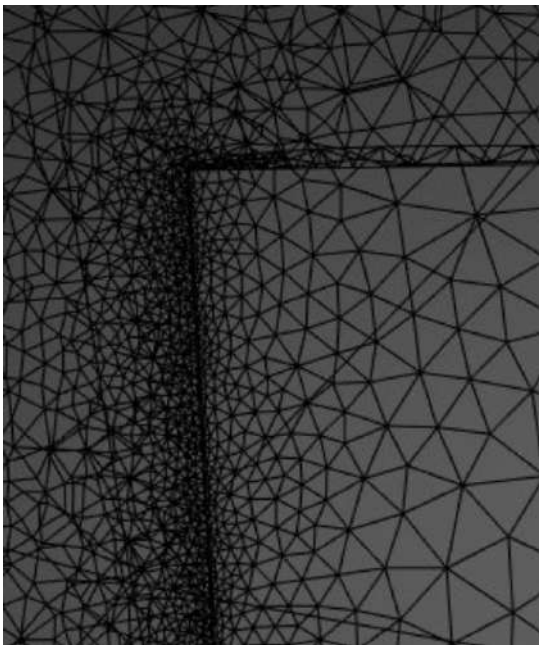
(a) Rover CAD (b) Simplified Rover CAD

Figure 3.1: Comparison of CAD Geometries

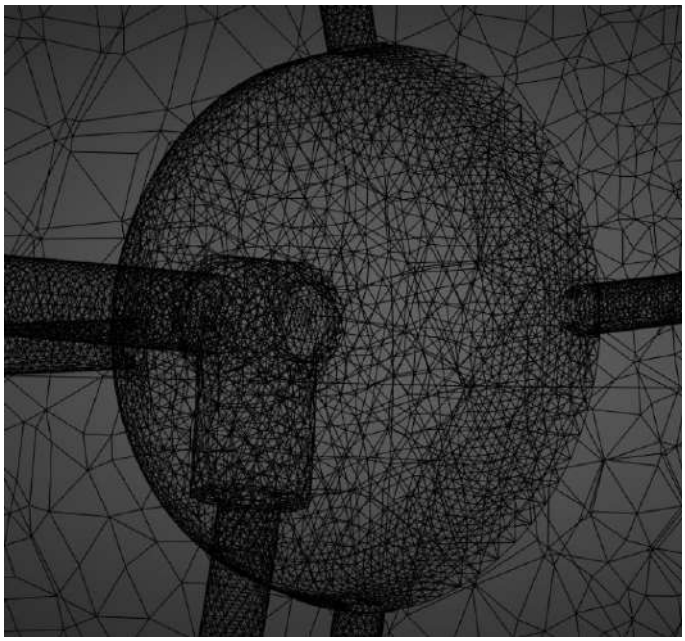
3.2.3 Mesh Study

An unstructured mesh was generated for the CFD simulations. As is standard practice in high-fidelity external aerodynamic simulations, finer mesh resolution was applied in regions with strong geometric curvature and expected high flow gradients, while coarser elements were used in the far-field domain to reduce computational cost.

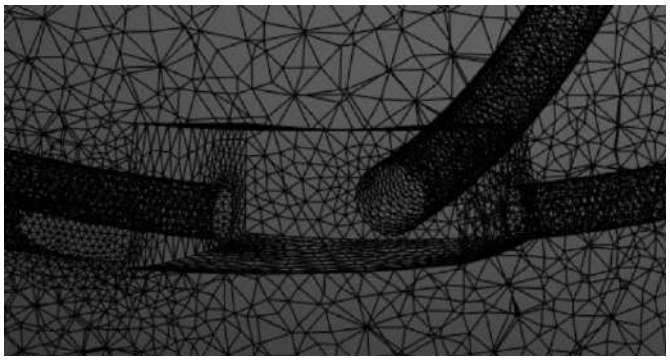
For each sail, as shown in Fig. 3.2b, the mesh followed conventional aerodynamic meshing strategies as mentioned in Costenoble et al. (2022). The highest element density was concentrated near the leading edge to accurately capture steep pressure gradients and potential



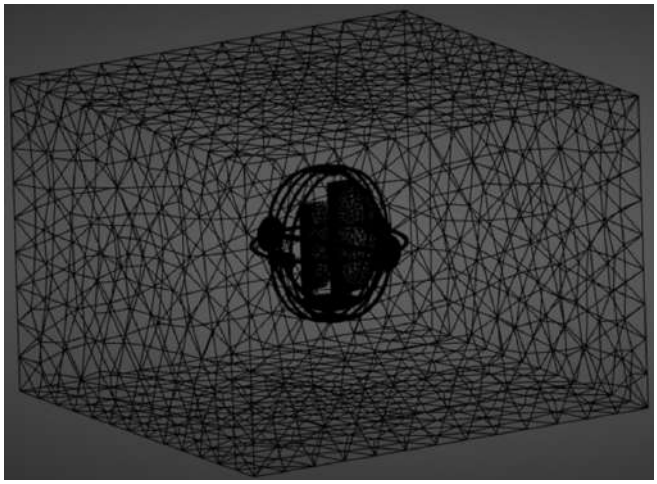
(a) Main Sail Leading Edge Mesh



(b) Side Plate Mesh



(c) Traction Hoop Connector Mesh



(d) CFD Rover Setup

Figure 3.2: Mesh details of the CFD model highlighting critical geometric regions.

flow separation. The mesh gradually coarsened toward the trailing edge where flow gradients are comparatively weaker. A curvature-based refinement strategy was employed with a minimum element size of 0.9 mm, ensuring adequate resolution of local geometric features and near-surface flow behavior.

Moving away from the sails into less geometrically complex regions, the mesh size progressively increased. In the far-field wind tunnel domain, the maximum element size reached approximately 0.18 m near the enclosure walls to limit computational expense while maintaining solution stability.

Other rover components were meshed using a similar refinement strategy. Regions with geometric discontinuities and structural junctions, such as the rectangular brackets where circular shell rods intersect (Fig. 3.2c), received localized mesh refinement to properly resolve flow interaction effects. The side hub region (Fig. 3.2d) followed the same refinement approach, with denser elements near sharp transitions and connection interfaces.

The final computational domain, shown in Fig. 3.2a, illustrates the gradual transition from dense near-body mesh to coarser far-field elements. Due to the use of an unstructured tetrahedral mesh, minor numerical overshoot near sharp airfoil features was anticipated, particularly around the NACA sections, as presented in Abobaker et al. (2021). However, the chosen refinement strategy ensured adequate resolution of key aerodynamic features while maintaining reasonable computational cost.

3.3 CFD Results

The results of the CFD simulations are presented in Fig. 3.3 and Fig. 3.4. The overall aerodynamic trends are consistent with those reported for finite-wing configurations, presented in Reca Luque et al. (2023); Sharma and Visbal (2019); Tao et al. (2021). For all yaw configurations, the lift force increases with increasing main-sail angle of attack. Drag also increases gradually with angle of attack, although at a slower rate compared to lift.

At the higher dynamic pressure (1.5 PSF), both lift and drag magnitudes increase proportionally, as expected from aerodynamic scaling. The drag curves exhibit a relatively mild slope with respect to angle of attack, partially due to the structural contributions of the rover cage and non-lifting components. These structural elements introduce baseline parasitic drag, resulting in a near-horizontal trend compared to the stronger lift variation.

The stall angles observed in the CFD simulations are higher than the typical 10°–12° stall range reported for isolated NACA 0020 airfoils as discussed in MAZUMDER (2024). This deviation is primarily attributed to the low aspect ratio sail geometry, coupled aerodynamic interaction between the main sail and jib, and flow disturbances generated by the surrounding

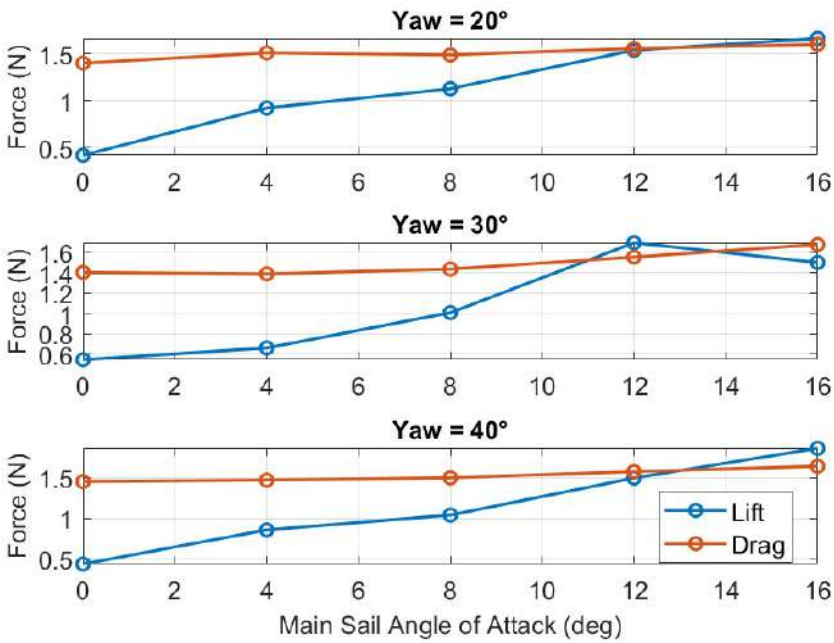


Figure 3.3: Simulation at 1 PSF

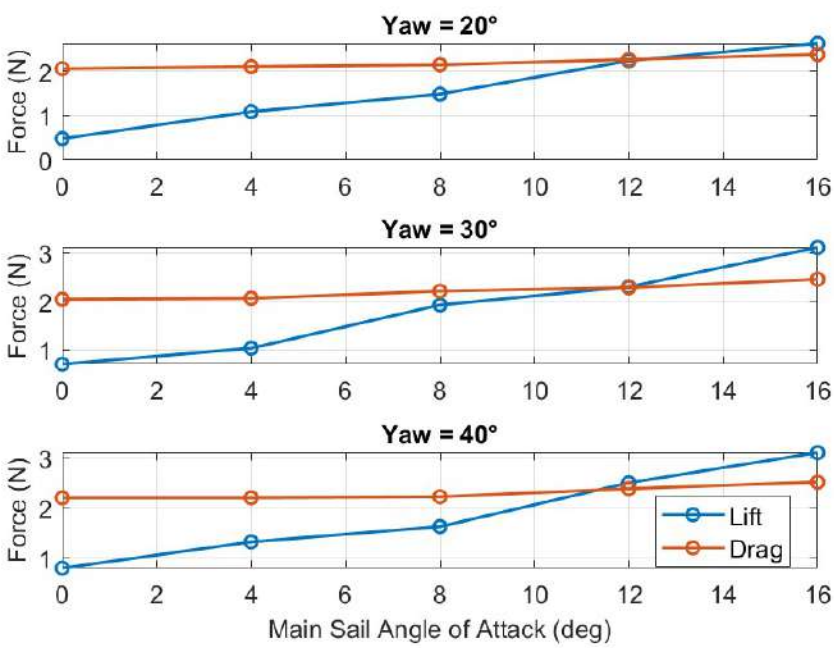
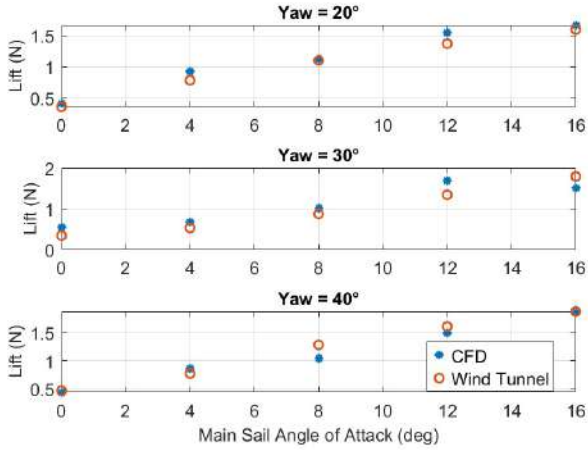


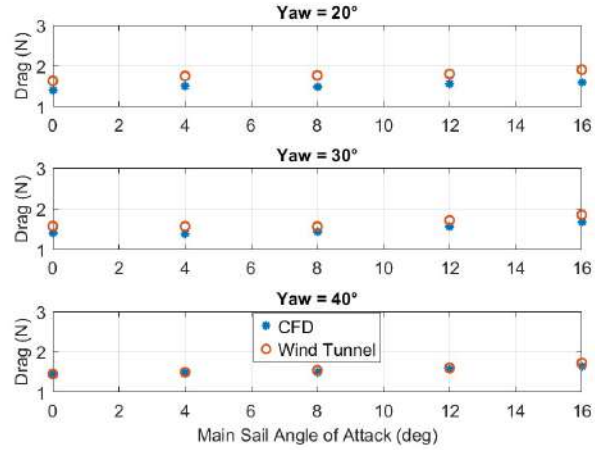
Figure 3.4: Simulation at 1.5 PSF

structural framework. These combined effects modify the pressure distribution and delay classical airfoil stall behavior.

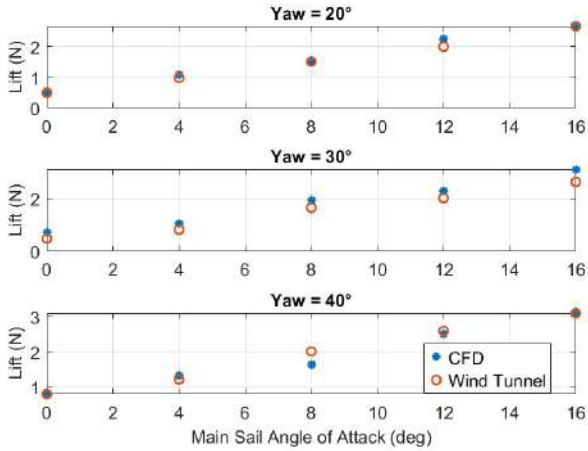
3.4 Comparison With Wind Tunnel Results



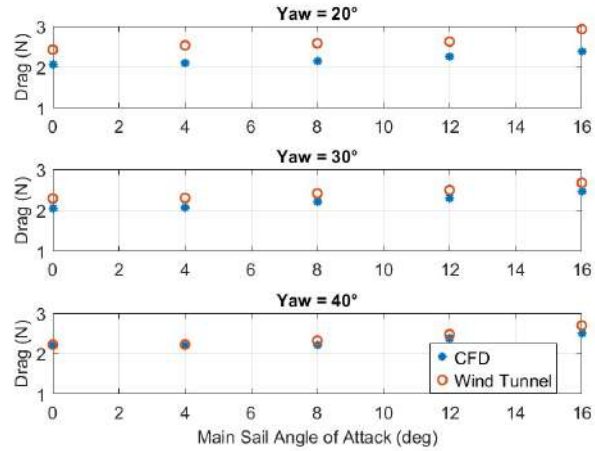
(a) Lift Comparison at 1 PSF



(b) Drag Comparison at 1 PSF



(c) Lift Comparison at 1.5 PSF



(d) Drag Comparison at 1.5 PSF

Figure 3.5: Comparison of CFD and Wind Tunnel Test 2 Results

The comparison between CFD simulation results and wind tunnel testing is presented in Fig. 4.12. Figures 3.5a and 3.5b show the lift and drag force comparisons at 1 PSF, while Figures 3.5c and 3.5d present the corresponding results at 1.5 PSF. Overall, both lift and drag forces exhibit similar trends between CFD simulations and experimental data across most test configurations. In general, the agreement improves for lift forces at larger yaw angles and higher wind speeds.

The largest discrepancies between the two methods are observed in the drag force results

at both 1 PSF (Fig.3.5b) and 1.5 PSF (Fig.3.5d). In these cases, the wind tunnel measurements consistently report higher drag than the CFD simulations. The closest agreement in drag occurs at a yaw angle of 40 degrees, while the greatest deviation is observed at a main sail angle of attack of 16 degrees. Across all yaw angles and wind speed conditions, the mean absolute error in lift and drag forces remains within approximately 10-12%.

Lift forces show the strongest similarity at lower yaw angles and higher main sail angle of attack. However, at a yaw angle of 40 degrees, Fig.3.5a and Fig.3.5c indicate a noticeable divergence at an 8-degree main sail angle. Unlike the drag forces, the CFD simulations often predict slightly higher lift compared to the wind tunnel measurements, as discussed in Abobaker et al. (2021).

3.4.1 Discussion of Differences

The methods discussed show correlation, and their errors can be identified for future mitigation. The proposed errors specifically focus on the CFD simulations. Factors contributing to inaccuracies in wind tunnels are explained by Fujii (2005). Out of those factors, the wind speed has been cross-verified by additional sensors during wind tunnel tests to keep it consistent.

A cause contributing to the error between the CFD and wind-tunnel experiments stems from approximations in the simulation model relative to the real one, as mentioned in 3.2.2. Further, the simulations used an unstructured mesh, which introduced some level of inaccuracy. A study in Abobaker et al. (2021) talks about how an unstructured mesh overestimates lift force compared to wind-tunnel tests and a structured mesh. However, the CFD analysis underpredicts the drag force compared to the wind-tunnel tests. This discrepancy mostly originates from how the traction hoop rotated in the wind tunnel, creating additional drag. The CFD simulations were run at two constant speeds, but the atmosphere could cause a small variation in the wind tunnel speed over the course of operations.

CHAPTER

4

NUMERICAL AERODYNAMIC MODEL

4.1 Introduction

For an early-stage design and parametric study, CFD often presents significant limitations. Some of them include high computation cost associated with mesh generation, solver convergence, and post-processing, making design optimization expensive, as explained in Rojas-Sola et al. (2016). In contrast, low-fidelity aerodynamic models based on the potential flow theory offer substantial advantages in terms of computational efficiency while still capturing the required forces with decent accuracy. One such numerical technique is the Weissinger method, an extension of the classical lifting-line theory. It discretizes lifting surfaces into bound vortices with trailing vortex sheets and enforces a no-flow boundary condition at control points along the surface, as explained in Weissinger (1947). This approach enables the prediction of lift distribution, induced drag, and aerodynamic coupling effects at a fraction of the computational cost of CFD. For configurations where flow separation is limited and angles of attack remain moderate, i.e., within the stall range, the Weissinger method has been shown to provide reliable accuracy, making it a go-to tool for parametric study or optimization.

The usefulness of the Weissinger method becomes particularly evident in multi-surface aerodynamic problems, such as tandem wings, canards, and multi-sail systems, where mutual interference plays a critical role in overall performance. By modeling the induced velocity

field from each lifting surface, the method naturally captures aerodynamic interaction effects, including upwash, downwash, and lift amplification due to proximity and alignment. These capabilities make the Weissinger method an attractive choice for analyzing the coupled aerodynamics of SSailOR's main and jib sail configuration, where understanding relative sail placement, trim angles, and spacing is essential for maximizing lift, upwind motion, and control authority.

To verify and validate the Weissinger-based implementation, XFLR5 is employed as an independent numerical tool. XFLR5 is a widely used aerodynamic analysis software that integrates lifting-line theory (LLT), vortex lattice methods (VLM), and panel methods for the analysis of airfoils, wings, and sail-like lifting surfaces, with a detailed study found in Deperrois (2013). Its versatility has led to extensive adoption in sail aerodynamics, low-speed aircraft design, unmanned aerial vehicles, and wind-driven vehicle research. XFLR5 allows rapid evaluation of lift, drag, and moment coefficients across a wide range of operating conditions and provides an effective reference for validating custom aerodynamic models. While XFLR5 itself is not a high-fidelity CFD solver, its implementation of vortex-based methods has been validated against experimental data and higher-fidelity simulations in numerous studies, demonstrating its suitability for conceptual design and comparative analysis, as shown in Communier et al. (2015).

Overall, the combination of the Weissinger method and XFLR5 offers a powerful and computationally efficient framework for aerodynamic analysis and comparison. This approach enables rapid exploration of a system that is in an initial design phase and provides physical insight into aerodynamic behavior, which is difficult to achieve using CFD alone during early-stage development. As such, low-fidelity CFD methods play a critical role in guiding the design and optimization of wind-propelled robotic platforms like SSailOR before transitioning to higher-fidelity simulations or experimental testing.

4.2 Weissinger Method For Sails

4.2.1 Methodology

In the Weissinger formulation, the lifting surface is discretized into a finite number of spanwise sections, each represented by a horseshoe vortex consisting of a bound vortex segment and trailing vortex legs extending downstream, shown in Fig. 4.1, taken from Şenipek (2017). To ensure consistency with thin airfoil theory, which predicts a lift-curve slope of 2π per radian (or 0.11 per degree), from Liu (2021), the bound vortex of each section is placed along the quarter chord line, while the corresponding control point is located at the three-quarter chord position.

At each control point, a zero normal-flow boundary condition is enforced, ensuring that the local flow remains tangent to the lifting surface.

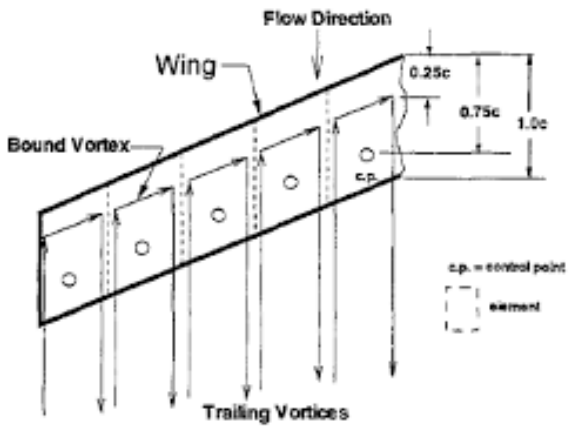


Figure 4.1: Weissinger Method Illustration

The normal flow velocity at a given control point consists of two contributions: the component of free stream velocity normal to the surface and the induced velocity generated by all horseshoe vortices in the system. Enforcing the zero normal-flow boundary condition requires the sum of these contributions to vanish at each control point.

$$V_{n,i} + (V_{\infty,n})_i = 0 \tag{4.1}$$

where $V_{n,i}$ is the induced normal velocity and $(V_{\infty,n})_i$ is the normal component of the freestream velocity at the ith control point. The freestream contribution depends on the local geometric orientation of the section and is expressed in terms of the geometric angle of attack, section twist, and zero-lift angle. The effective geometric angle of attack accounts for sweep through a trigonometric projection of the freestream velocity onto the section normal.

The induced velocity contribution in (4.1) is computed using the Biot-Savart law, accounting for the influence of all bound and trailing vortex segments on each control point**?**. To solve for the unknown circulation strengths, an aerodynamic influence coefficient (AIC) matrix is constructed. Each entry of the matrix represents the induced normal velocity at a given control point due to a unit-strength vortex at another section. The resulting linear system takes the form:

$$[F]\{\Gamma\} = \{V_n\} \tag{4.2}$$

where [F] is the N x N AIC matrix, where N is the number of horseshoe vortices, Γ is the vector of

unknown circulations and (V_n) is the vector of freestream normal-velocity components. Figure 4.2 illustrates the concept of circulation about a lifting surface, where the bound vortex along the span induces downwash and generates lift. Solving equation (4.2) yields the circulation distribution along the span.

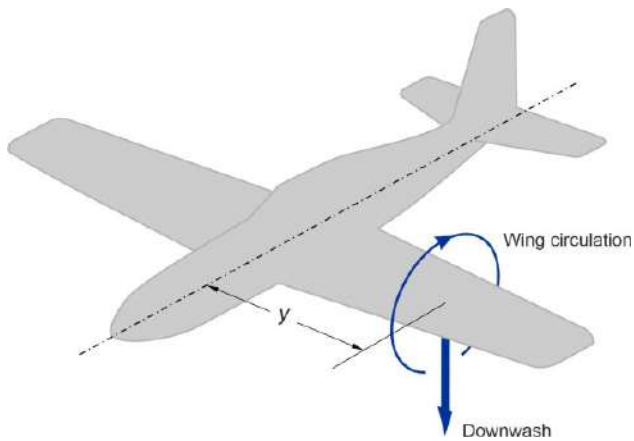


Figure 4.2: Circulation Around a Lifting Surface

Once the circulation distribution is obtained, sectional lift is computed using the Kutta-Joukowski theorem, discussed in Bai and Wu (2014), assuming that induced velocity components in the freestream direction are small relative to the freestream magnitude. The elemental lift for each section is given by:

$$dL_i = \rho V_\infty \Gamma_i \, ds_i \tag{4.3}$$

where ds_i is the spanwise length of the section. Using equation (4.3), the sectional lift coefficient is then expressed as:

$$C_{l,i} = \frac{2}{c_i}\left(\frac{\Gamma_i}{V_\infty}\right) \tag{4.4}$$

And the total lift coefficient is obtained by integrating the circulation distribution across the span:

$$C_L = \frac{2}{S_{\text{ref}}} \sum_{i=1}^{N} \left(\frac{\Gamma_i}{V_\infty}\right) \cos(\theta_i) \, ds_i \tag{4.5}$$

where θ_i is the local dihedral angle. The induced drag is computed from the product of

circulation and downwash at each section. The induced drag coefficient is expressed as:

$$C_{D_i} = \frac{2}{S_{\text{ref}}} \sum_{i=1}^{N} \left(\frac{w_i}{V_\infty} \right) \left(\frac{\Gamma_i}{V_\infty} \right) d\, s_i \tag{4.6}$$

where W_i represents the normal downwash velocity at the section and is calculated using:

$$[W] = [F_m]\{\Gamma\} \tag{4.7}$$

where $[F_m]$ is the N x N AIC matrix calculated at the mid-point instead of the control point, as discussed in D'Angelo (2021)

In this work, the Weissinger method provides an efficient framework for modeling the aerodynamic interaction between multiple lifting surfaces.

4.2.2 Implementation of Weissinger Method

A schematic of the discretized sail configuration, including bound vortices, control points, and trailing vortices, is shown in Fig 4.3. To apply the Weissinger method to the SSailOR's dual-sail system, the formulation was implemented in a modular manner, divided into 4 primary stages: 1. Initialization, 2. Geometric data generation, 3. AIC matrix assembly, 4. Force computation.

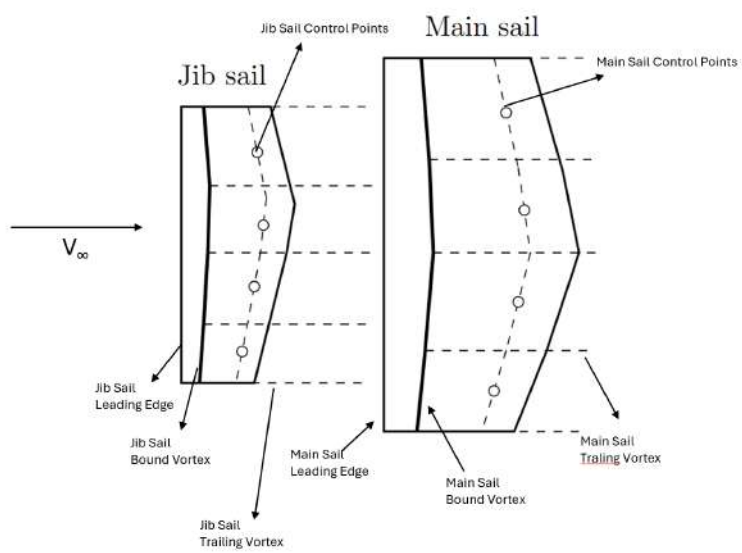


Figure 4.3: Weissinger Model Diagram For Sails (illustration only)

Initialization

The first stage of the implementation consists of reading a structured input file that defines the global aerodynamic parameters and the geometric description of each sail. This file-based approach allows rapid modification of sail geometry and operating conditions without altering the solver logic, making it particularly suitable for parametric studies and design optimization.

At the global level, the input file specifies reference quantities such as the reference area S_{ref}, reference span B_{ref}, reference chord C_{ref}, and symmetry options. The lifting surfaces are then defined as a collection of patches, where each patch represents a spanwise segment of either the main sail or jib sail. For each patch, the number of spanwise panels and the geometric endpoints are specified. These endpoints define the local chord length, zero-lift angle, and twist, enabling the modeling of nonuniform sail geometry.

Finally, the operating condition is defined through the global angle of attack. This structure allows the same framework to be used for multiple yaw orientations by rotating the sail geometry relative to the freestream direction. The reading inputs table used for this case is shown in Table 4.1. The definitions of the parameters used in Table 4.1 are listed below and are shown schematically in Fig. 4.4 for clarity.

a = distance from jib sail LE to the axis of rotation of jib sail

b= distance from the jib sail axis of rotation to the main sail axis of rotation

c = distance from main sail LE to the axis of rotation of main sail

$\boldsymbol{\theta}_j = \psi - \alpha_j$

Table 4.1: Input Parameters and Patch Geometry for Weissinger-Based Dual-Sail Model

Parameter	**Value**	**Units**
Symmetry	None	–
Number of patches, n_{patch}	4	–
Panels per patch, n_{pan}	50 (each patch)	–
Patch 1 (Jib) quarter-chord endpoints	$(x, y, z, c) = (0, 0, 0, 73)$ to $(0, 110, 0, 100)$	mm
Patch 2 (Jib) quarter-chord endpoints	$(x, y, z, c) = (0, 110, 0, 100)$ to $(0, 220, 0, 73)$	mm
Patch 3 (Main) quarter-chord endpoints	$(x, y, z, c) = (a + b\cos\theta_j, 0, b\sin\theta_j, 82)$ to $(a + b\cos\theta_j, 143, b\sin\theta_j, 160)$	mm
Patch 4 (Main) quarter-chord endpoints	$(x, y, z, c) = (a + b\cos\theta_j, 143, b\sin\theta_j, 160)$ to $(a + b\cos\theta_j, 286, b\sin\theta_j, 82)$	mm

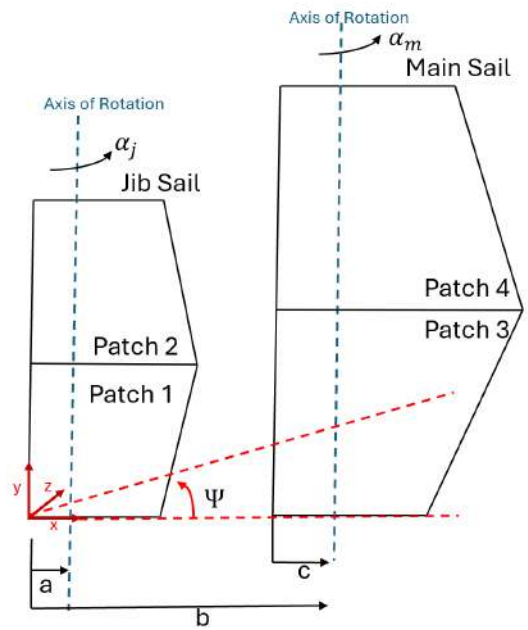


Figure 4.4: Geometry of Sails for Weissinger (Isometric View)

Geometric Data Generation

Once the input data are read, the geometry of each sail patch is processed to generate the Weissinger vortex system. Each patch is discretized into a prescribed number of spanwise panels. For each panel, the following parameters are computed:

- Quarter-chord points define the location of the bound vortices
- Three-quarter-chord points, which serve as control points
- Panel spanwise length and orientation

The bound vortex for each panel is placed along the quarter-chord line, extending between the panel's spanwise endpoints. From each bound vortex endpoint, a trailing vortex leg is extended downstream in the direction of the freestream velocity. This formulation naturally allows the induced velocity field from the jib sail to influence the main sail, and vice versa, enabling accurate modeling of coupled aerodynamics. All parameters for each panel are computed in a global reference frame, ensuring that changes in yaw angle are consistently reflected in the induced velocity calculations.

AIC Matrix Assembly

The third stage involves constructing the aerodynamic influence coefficient (AIC) matrix. Each element of the AIC matrix represents the induced normal velocity at a given control point due to

a unit-strength horseshoe vortex located on another panel. The induced velocity contributions from the bound vortex segment and both trailing vortex legs are computed using the Biot–Savart law. These induced velocities are projected onto the local surface normal at each control point. The freestream normal velocity component is then added to enforce the zero normal-flow boundary condition. Assembling these contributions for all control points and vortex elements yields a square linear system of the form expressed in equation 4.2. Solving this system provides the circulation distribution across both sails simultaneously, inherently capturing aerodynamic coupling effects.

Force Computation

In the final stage, the circulation distribution obtained from the linear system is used to compute aerodynamic forces. Sectional lift is calculated using the Kutta–Joukowski theorem (equation 4.3), assuming negligible induced velocity components in the freestream direction. The lift contributions from all panels are then integrated to obtain the total lift coefficient (equation 4.5). Induced drag is computed from the product of circulation and local downwash at each panel, allowing assessment of drag penalties associated with sail interaction and trim changes (equation 4.6). The resulting lift and drag forces are resolved relative to the freestream direction, enabling direct comparison with XFLR5 output and future experimental measurements.

4.2.3 Weissinger Method Results

The aerodynamic forces predicted using the Weissinger Vortex Lattice Method (VLM) are presented in this section. The numerical model was developed to estimate lift and drag forces generated by the dual-sail configuration across varying main sail angles of attack and yaw angles. Results are shown for dynamic pressures of 1 PSF and 1.5 PSF to enable direct comparison with wind tunnel measurements.

Figures 4.5 and 4.6 present the lift and drag forces predicted by the Weissinger method for yaw angles of 20°, 30°, and 40°. As expected, lift increases approximately linearly with increasing main sail angle of attack for all yaw configurations, consistent with thin airfoil and VLM assumptions. The 40° yaw configuration produces the highest lift magnitudes due to increased effective angle between the apparent wind and sail surface.

Drag values remain significantly smaller than lift and exhibit mild growth with angle of attack. Since the Weissinger method is an inviscid formulation, drag predictions primarily represent induced drag contributions, with viscous effects not explicitly captured. Consequently, drag magnitudes are lower than those observed in wind tunnel testing.

Comparison across dynamic pressures shows proportional scaling of both lift and drag

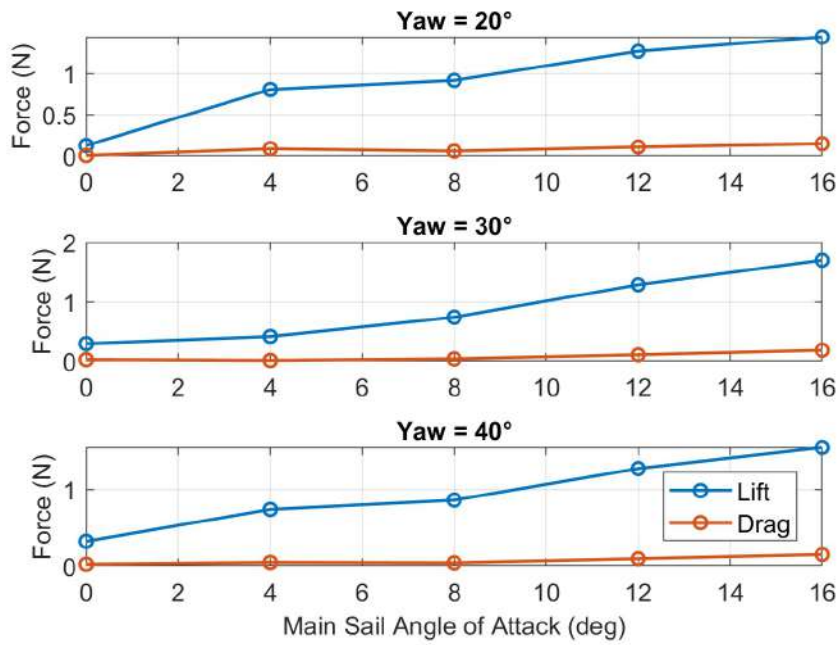


Figure 4.5: Weissinger At 1 PSF

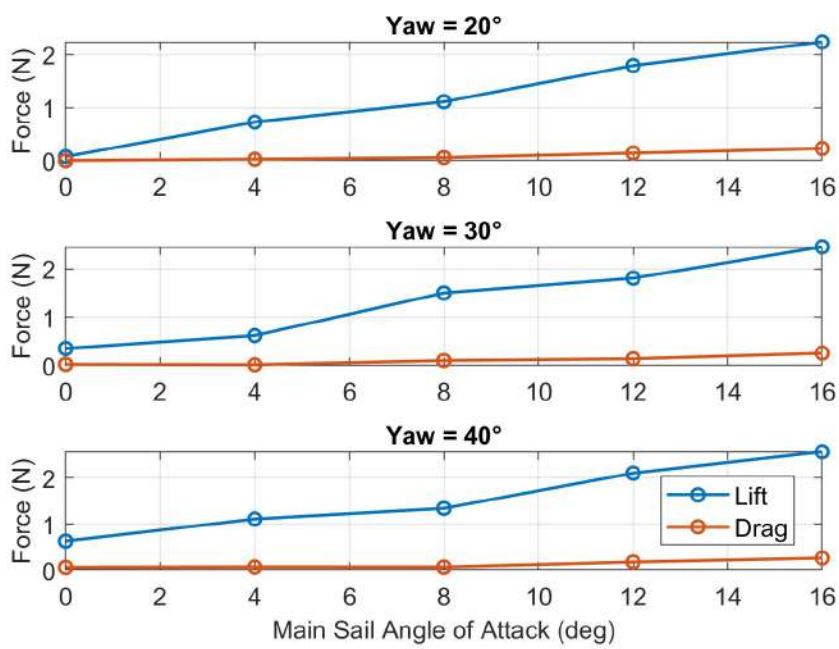


Figure 4.6: Weissinger At 1.5 PSF

forces, confirming consistency of the numerical model with aerodynamic force scaling principles.

4.3 XFLR For Sails

4.3.1 Methodology

XFLR5 is a program that puts together other software, specifically XFOIL and PROFOIL, for the design and simulation of airfoils, wings, and aircraft. This program is supported by mathematical methods, including the Vortex Lattice Method, Lifting Line Theory, and the 3D Panel Method, to model the macroscopic effects and to analyze behavior of aircraft, as found in Loya et al. (2018), Güzelbey et al. (2018), Akter et al. (2024), Turon Suñer (2025). XFLR5 solves for aerodynamic performance at low Reynolds numbers, capable of gathering lift, drag, pressure, and pitching moment. XFLR5 is a useful tool for analyzing 2D airfoils, finite-wings, and airplanes.

The sails simulated are in inviscid flow. Inviscid flow disregards the friction effects in the fluid and at the boundary. In XFLR5, the airfoil is first made in XFOIL. XFOIL analyzes inviscid fluid in the following procedure, further explained in Güzelbey et al. (2018), Drela (1989). The aerodynamic effect of an airfoil is determined by solving along the airfoil using "nodes." A 2D streamline function for a given geometry of a vortex sheet on the airfoil with a source sheet on the wake and surface is expressed.

$$\Psi(x,y) = u_\infty - v_\infty + \frac{1}{2\pi}\int \gamma(s)\ln r(s;x,y)\,ds + \frac{1}{2\pi}\int \sigma(s)\theta(s;x,y)\,ds$$

For a constant value of Ψ_0 across several nodes, this stream function can be solved into a linear system.

$$\sum_{j=1}^{N} a_{i,j}\gamma_j - \Psi_0 = -u_\infty\, y_i + v_\infty\, x_i - \sum_{j=1}^{N} b_{i,j}\,\sigma_{0,j}, \qquad 1 \le i \le N$$

The additional coefficients $a_{i,j}$ and $b_{i,j}$ can be determined from the streamline function when the panel nodes x_i and y_j and the wake nodes are known. Applying the Kutta condition simplifies the expression as shown below.

$$\gamma_1 + \gamma_N = 0$$

Since the flow inside the airfoil is stagnant, the surface vorticity and velocity are equal. Using Bernoulli's equation, the pressure coefficient can be expressed in terms of the surface vorticity. From here, the other elements of lift and drag are straightforward to solve.

$$C_p = \frac{\gamma}{q_\infty}$$

4.3.2 Application of XFLR5 to the dual-sail configuration

To verify the Weissinger method results, an independent aerodynamic analysis was performed using XFLR5. Since XFLR5 is formulated around aircraft lifting surfaces, the SSailOR system was mapped onto an equivalent configuration by modeling the jib sail as the main wing and the main sail as an elevator. This approach enables the use of XFLR5's built-in vortex lattice and lifting-line solvers while preserving the relative positioning and aerodynamic interaction between the two sails.

For all yaw cases, the jib sail (main wing) was placed at the reference origin, with its leading edge located at x = 0 and z = 0. The main sail (elevator) was positioned downstream and offset relative to the jib sail. The geometric offset between the leading edges of the jib and main sails was maintained at a fixed distance of 0.12 m, corresponding to the physical separation in the SSailOR configuration.

20 Degree Yaw

For the 20-degree yaw case, the jib sail (main wing) was fixed at the reference origin with its leading edge located at x=0 and z=0. The corresponding main sail (elevator) location was computed using simple trigonometric relationships:

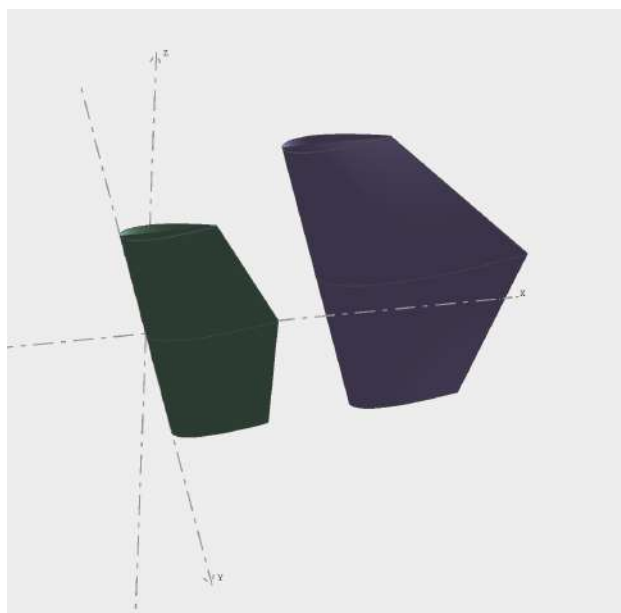


Figure 4.7: SSailOR at 20 Degree Yaw in XFLR5

$$z = 0.12\tan(20^\circ) \tag{4.8}$$

$$x = \frac{0.12}{cos(20^\circ)} \tag{4.9}$$

This configuration preserves the correct projected spacing between the sails when exposed to a 20° yawed freestream, as illustrated in Fig. 4.7

30 degree yaw

For the 30-degree yaw case, the jib sail remained fixed at x=0, z=0, while the main sail position was adjusted to reflect the increased yaw angle. Using the same geometric separation of 0.12 m, the elevator offsets were determined as:

$$z = 0.12\tan(30^\circ) \tag{4.10}$$

$$x = \frac{0.12}{cos(30^\circ)} \tag{4.11}$$

As shown in Fig. 4.8, this results in a larger vertical offset and downstream placement of the main sail, increasing the effective interaction between 2 surfaces under 30° yawed freestream

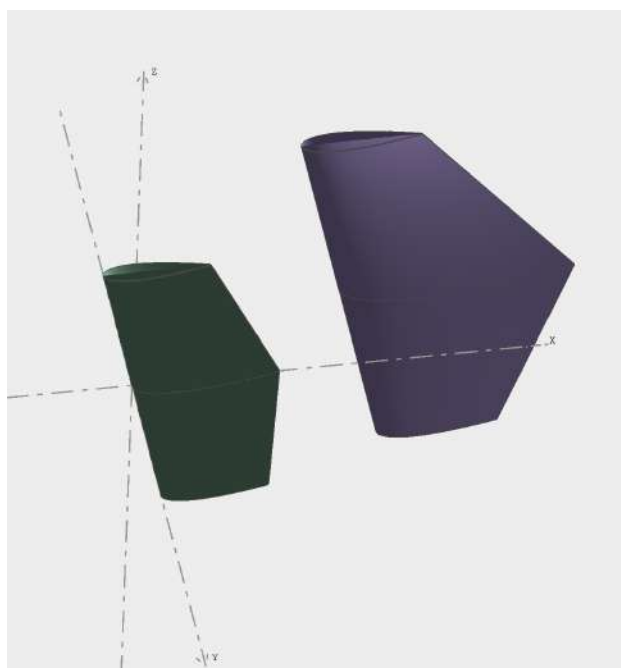


Figure 4.8: SSailOR at 30 Degree Yaw in XFLR5

40 dergree yaw

For the 40-degree yaw case, the same modeling approach was applied, with jib sail fixed at the origin and the main sail positioned using:

$$z = 0.12 \tan(40^\circ) \tag{4.12}$$

$$x = \frac{0.12}{cos(40^\circ)} \tag{4.13}$$

This configuration produces the largest geometric offset between the sails, as illustrated in Fig.4.9, and represents the most aggressive yaw condition examined in this study. Despite the increased separation, the mapping preserves geometric consistency with the Weissinger-based model and enables meaningful aerodynamic comparison.

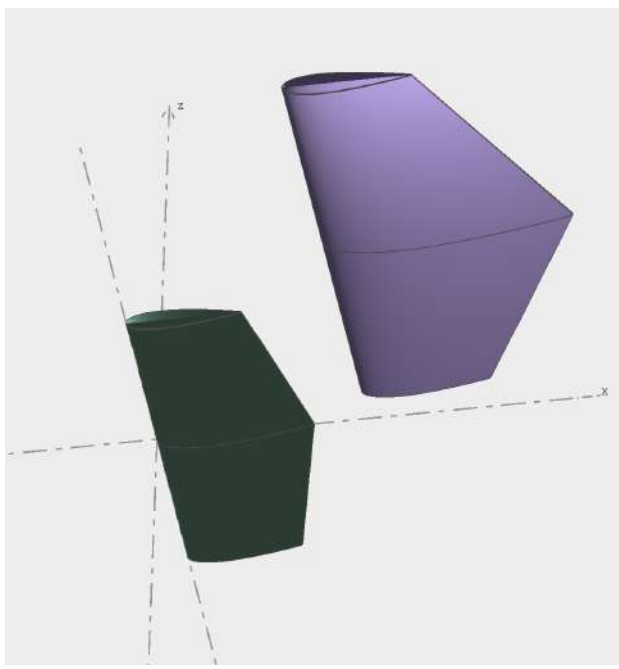


Figure 4.9: SSailOR at 40 Degree Yaw in XFLR5

4.3.3 XFLR5 Results

The aerodynamic forces computed by XFLR5 for the dual-sail configuration are shown in Fig. 4.10 and Fig. 4.11 for yaw angles of 20°, 30°, and 40°. Across all yaw cases, lift increases with the main sail angle of attack, while the induced drag remains significantly smaller and grows

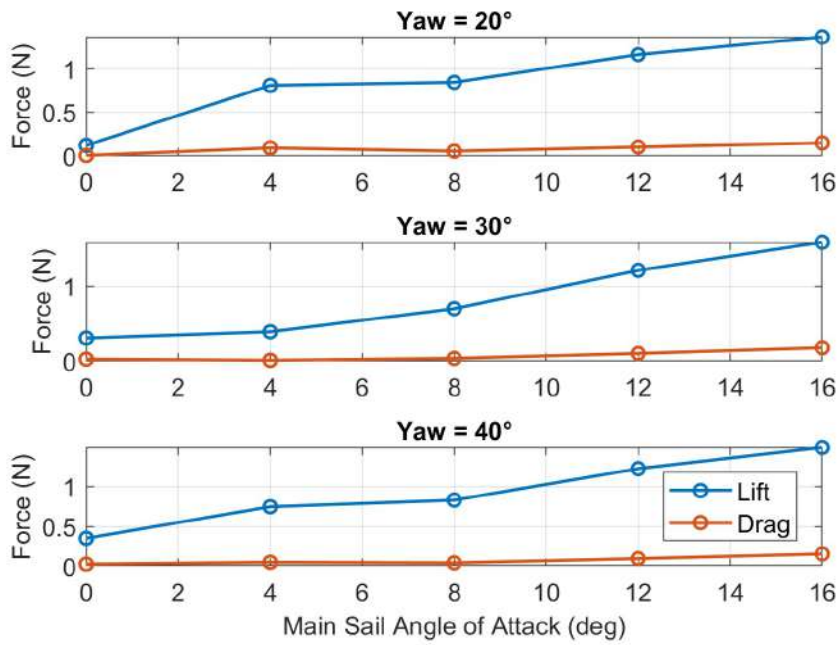


Figure 4.10: XFLR5 At 1 PSF

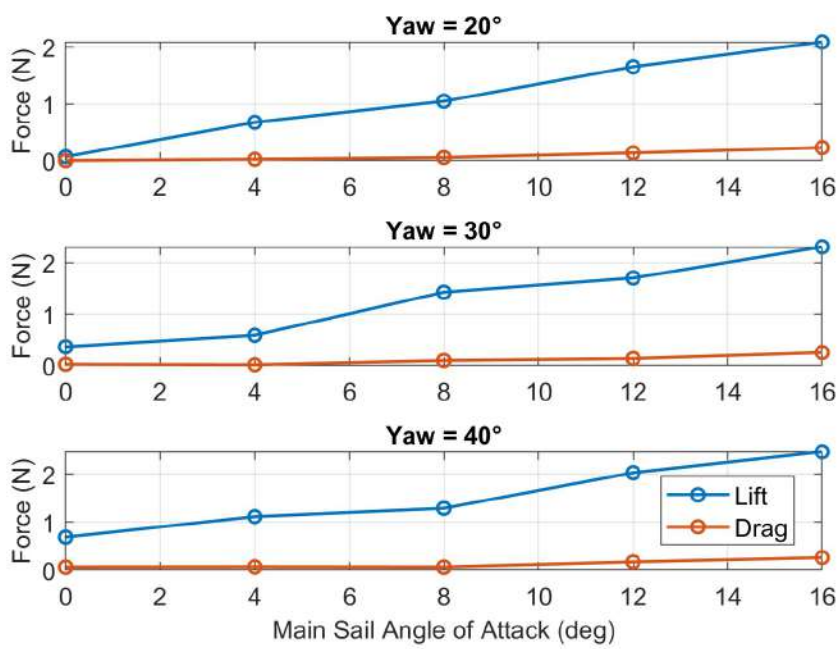
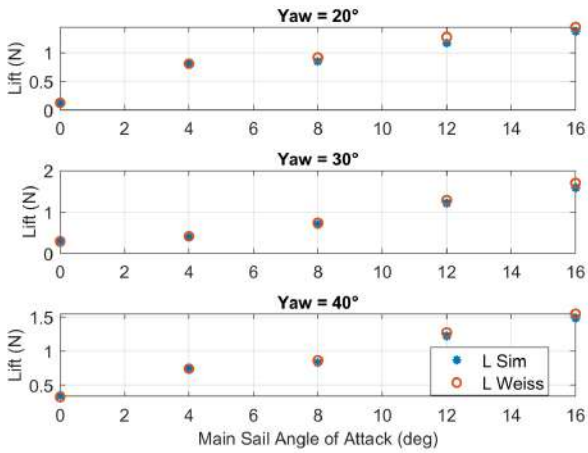


Figure 4.11: XFLR5 At 1.5 PSF

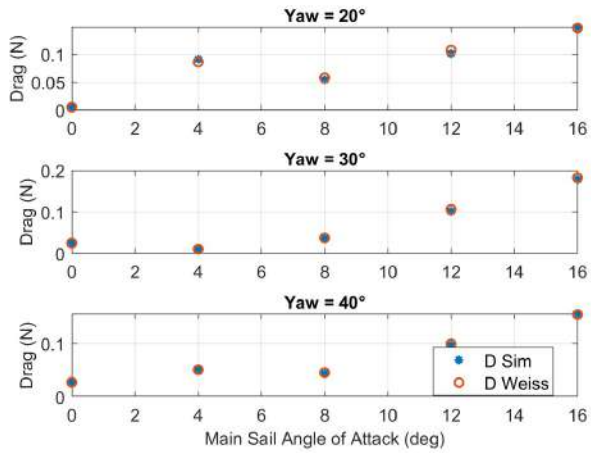
gradually. Higher yaw angles increase lift for a given sail trim, consistent with the increased effective flow incidence and enhanced interaction between the jib (main wing) and the main sail (elevator). The lift trends remain smooth and largely linear over the examined angle-of-attack

range, indicating stable numerical behavior of the XFLR5 vortex-based solver.

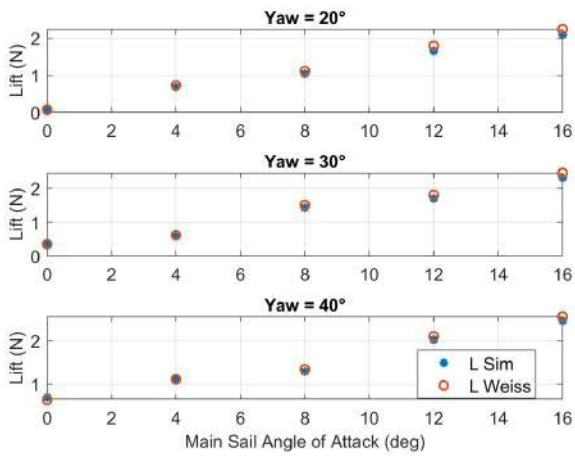
4.4 Comparison



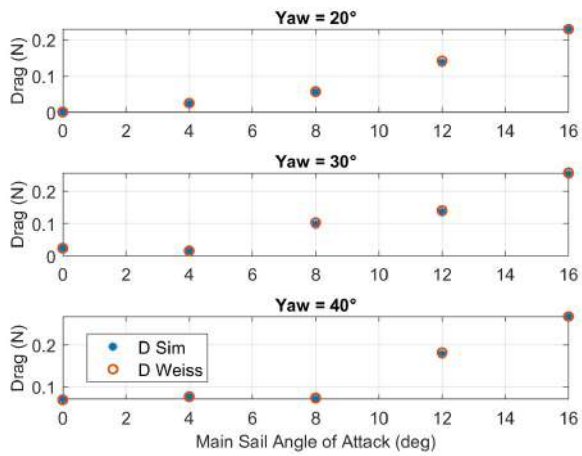
(a) Lift Comparison at 1 PSF



(b) Drag Comparison at 1 PSF



(c) Lift Comparison at 1.5 PSF



(d) Drag Comparison at 1.5 PSF

Figure 4.12: Comparison of Weissinger and XFLR5 Results

Figures 4.12a, 4.12b, 4.12c, 4.12d compare the lift and induced drag forces obtained from the Weissinger method and XFLR5 across yaw angles of 20°, 30°, and 40° and wind speeds of 1 PSF and 1.5 PSF. Overall, the two methods show strong agreement in both lift and drag trends over the examined range of main sail angles of attack and other parameters. Lift forces computed from the Weissinger model closely match XFLR5 Lift forces for all yaw cases, with

minor deviations appearing at higher angles of attack, where increased induced effects and modeling assumptions begin to influence the results.

Drag forces also show good agreement, with both methods capturing the gradual increase in drag with angle of attack and yaw angle. Small discrepancies are observed at higher yaw angles and larger sail trims, which can be attributed to differences in how induced drag and multi-surface interaction are modeled in the two approaches. Despite these differences, the Weissinger method reliably reproduces the overall aerodynamic behavior predicted by XFLR5, validating its use as a fast and effective reduced-order tool for parametric analysis and early-stage design of the dual-sail configuration.

While the Weissinger method and XFLR5 methods show consistent aerodynamic trends, the remaining differences highlight the limitations of low-fidelity models for interacting lifting surfaces. The Weissinger method assumes inviscid flow and simplified wake behavior. This limits its ability to fully capture nonlinear effects that become more dominant at higher yaw and sail angles. These assumptions primarily influence induced drag prediction and the distribution of circulation across the sails.

Additional discrepancies arise from the geometric abstraction required in XFLR5, where the dual-sail system is mapped onto a wing-elevator configuration. Although this approach preserves relative positioning and interaction effects, it introduces simplifications in wake alignment and surface coupling that affect force computation. Despite these limitations, the observed differences remain well within expected bounds for low-fidelity models, making Weissinger a useful tool for rapid parametric studies and early-stage design, while motivating higher-fidelity simulations and experimental validation in future work.

4.5 Conclusion

This chapter presents an aerodynamic analysis of a dual-sail configuration for the SSailOR using the Weissinger method and validated using XFLR5. The results demonstrate that the Weissinger model captures the dominant aerodynamic trends associated with sail trim and yaw, showing strong agreement with XFLR5 across a broad operating range. These findings confirm the suitability of the Weissinger method as a fast and effective tool for early-stage design and parametric exploration of wind-propelled lift-based robotic platforms. Future work will focus on higher-fidelity CFD simulations and experimental validation to further refine the aerodynamic model and quantify effects beyond the scope of reduced-order methods.

CHAPTER

5

SSAILOR DYNAMIC MODEL

5.1 Introduction

A dynamic model of SSailOR is essential to understand and predict the rover's motion under wind-driven propulsion. Unlike conventional wheeled rovers, SSailOR relies on coupled aerodynamic forces generated by a dual-sail configuration and ground interaction forces to produce motion. The dynamic behavior of the system is therefore governed by the interactions among aerodynamic forces, internal mass distribution, rotational kinematics, and ground-contact constraints.

The development of a high-fidelity dynamic model enables simulation of rover motion under varying wind speeds, yaw orientations, sail angles of attack, and rotational speeds. Such a model is critical for evaluating performance metrics, including propulsion efficiency, stability, and achievable upwind motion. Furthermore, the dynamic model provides the foundation for control system design, trajectory planning, and closed-loop sail actuation strategies.

Due to the geometric complexity of SSailOR and the coupled interactions between the spherical shell, internal hull, and aerodynamic surfaces, simplified point-mass approximations are insufficient. A multibody formulation is required to accurately represent the system's degrees of freedom and constraint relationships. In this work, Kane's method is employed to derive the equations of motion. Kane's method offers a systematic and computationally

efficient approach for modeling constrained multibody systems and is particularly well-suited for systems with multiple generalized coordinates and nonholonomic constraints.

The following sections describe the selection of generalized coordinates, derivation of kinematic relationships, formulation of generalized active and inertial forces, and the resulting equations of motion. The derived model is subsequently implemented for numerical simulation to evaluate SSailOR's dynamic response under representative operating conditions.

5.2 Kane's Method

Kane's method is a systematic formulation for deriving the equations of motion of multibody systems using generalized speeds, partial velocities, and partial angular velocities. Unlike Newton–Euler or Lagrange formulations, which typically form r second-order differential equations in the generalized coordinates, Kane's method produces $2r$ first-order differential equations in generalized coordinates and generalized speeds. This formulation results in computationally efficient and compact equations of motion, making it well-suited for complex, constrained systems such as SSailOR.

The system is described using $q_1, \ldots, q_r$ generalized coordinates and $\dot{q}_1, \ldots, \dot{q}_r$ generalized velocities. Kane's method introduces a set of generalized speeds $u_1, \ldots, u_r$, which are defined as linear combinations of the generalized velocities,

$$[u] = [Y][\dot{q}] - [Z] \tag{5.1}$$

where $[Y]$ and $[Z]$ are configuration-dependent matrices and vectors. The only requirement for the generalized speeds is that they must uniquely define the generalized velocities. Therefore, the matrix $[Y]$ must be invertible, allowing the generalized velocities to be expressed as

$$[\dot{q}] = [W][u] + [Z] \tag{5.2}$$

where $[W] = [Y]^{-1}$.

Using generalized speeds, the equations of motion are written in terms of generalized active forces F_r and generalized inertia forces F_r^*, resulting in Kane's equation,

$$F_r + F_r^* = 0 \tag{5.3}$$

for each generalized speed u_r.

The generalized active force is defined as

$$F_r = \sum_{k=1}^{N_R} \left(\mathbf{F}_k \cdot \bar{o}\mathbf{v}_{cm_k,r} + \boldsymbol{\tau}_k \cdot \bar{o}\boldsymbol{\omega}_{B_k,r} \right) + \sum_{l=1}^{N_P} \mathbf{f}_l \cdot \bar{o}\mathbf{v}_{m_l,r} \tag{5.4}$$

where N_R is the number of rigid bodies, N_P is the number of particles, $\mathbf{F}_k$ and $\boldsymbol{\tau}_k$ are the resultant force and torque acting on the kth rigid body, and $\mathbf{f}_l$ is the force acting on the lth particle. The terms $\bar{O}\mathbf{v}_{cm_k,r}$ and $\bar{O}\boldsymbol{\omega}_{B_k,r}$ are the rth partial velocity and partial angular velocity, respectively.

The generalized inertia force is given by

$$\begin{aligned} F_r^* = -\sum_{k=1}^{N_R} \Big(& m_k\, \bar{o}\mathbf{a}_{cm_k/O} \cdot \bar{o}\mathbf{v}_{cm_k,r} \\ & + \left(\tilde{I}_{cm_k} \cdot \bar{o}\boldsymbol{\alpha}_{B_k} + \bar{o}\boldsymbol{\omega}_{B_k} \times (\tilde{I}_{cm_k} \cdot \bar{o}\boldsymbol{\omega}_{B_k}) \right) \cdot \bar{o}\boldsymbol{\omega}_{B_k,r} - \sum_{l=1}^{N_P} m_l\, \bar{o}\mathbf{a}_{m_l/O} \cdot \bar{o}\mathbf{v}_{m_l,r} \end{aligned} \tag{5.5}$$

For a rigid body, the velocity and angular velocity can be expressed in terms of generalized speeds as

$$\bar{o}\mathbf{v}_{cm_k/O} = \sum_{r=1}^{n} \left(\bar{o}\mathbf{v}_{cm_k,r}\, u_r \right) + \bar{o}\mathbf{v}_{cm_k,t} \tag{5.6}$$

$$\bar{o}\boldsymbol{\omega}_{B_k} = \sum_{r=1}^{n} \left(\bar{o}\boldsymbol{\omega}_{B_k,r}\, u_r \right) + \bar{o}\boldsymbol{\omega}_{B_k,t} \tag{5.7}$$

where $\bar{o}\mathbf{v}_{cm_k,r}$ and $\bar{o}\boldsymbol{\omega}_{B_k,r}$ are the partial velocity and partial angular velocity associated with generalized speed u_r. These are obtained by

$$\bar{o}\mathbf{v}_{cm_k,r} = \frac{\partial}{\partial u_r} \left(\bar{o}\mathbf{v}_{cm_k/O} \right) \tag{5.8}$$

$$\bar{o}\boldsymbol{\omega}_{B_k,r} = \frac{\partial}{\partial u_r} \left(\bar{o}\boldsymbol{\omega}_{B_k} \right) \tag{5.9}$$

Similarly, for a particle,

$$\bar{o}\mathbf{v}_{m_l/O} = \sum_{r=1}^{n} \left(\bar{o}\mathbf{v}_{m_l,r}\, u_r \right) + \bar{o}\mathbf{v}_{m_l,t} \tag{5.10}$$

with partial velocities computed as

$$\bar{o}\mathbf{v}_{m_l,r} = \frac{\partial}{\partial u_r} \left(\bar{o}\mathbf{v}_{m_l/O} \right) \tag{5.11}$$

This formulation provides a structured framework for deriving the equations of motion of SSailOR while efficiently handling multibody coupling and constraint relationships.

5.3 System Definition

5.3.1 Defining Frames

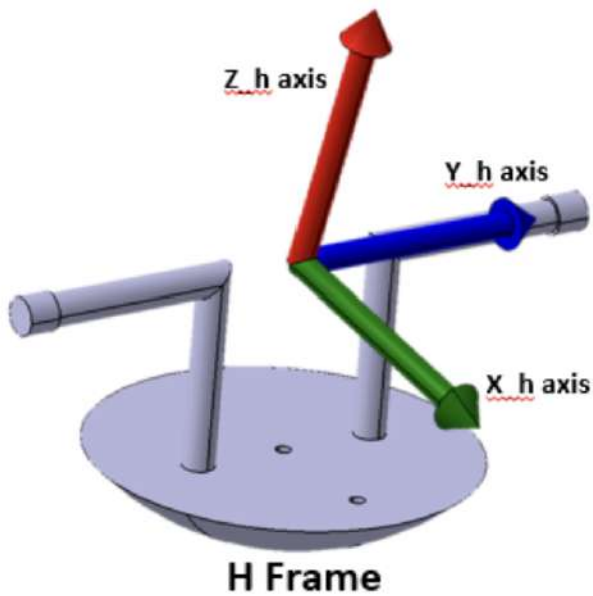


Figure 5.1: Hull Frame

The SSailOR system is modeled as a coupled multibody system consisting of two rigid bodies: the outer spherical shell and the internal hull assembly. The configuration of these bodies is illustrated in Fig. 5.1 and Fig. 5.2.

Inertial Reference Frame

An inertial reference frame $O = \{\mathbf{i}_O, \mathbf{j}_O, \mathbf{k}_O\}$ is fixed to the ground. The unit vectors $\mathbf{i}_O$ and $\mathbf{j}_O$ lie in the horizontal plane, while $\mathbf{k}_O$ is directed vertically upward. All position, velocity, and acceleration quantities are ultimately expressed relative to this frame.

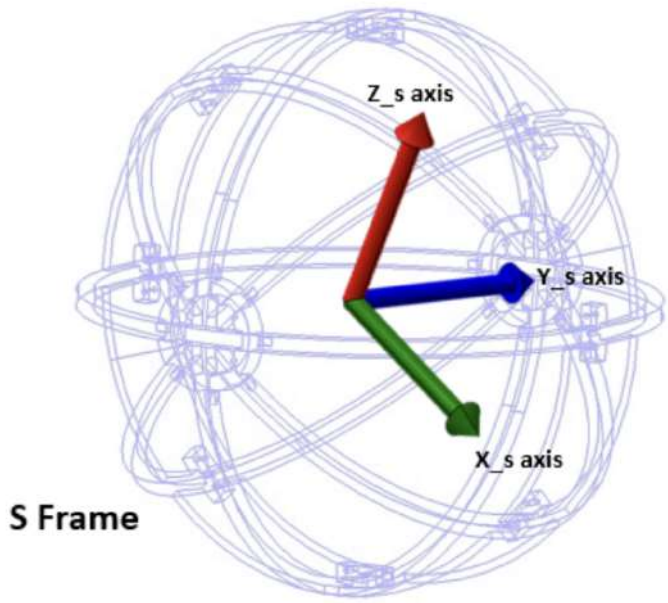


Figure 5.2: Sphere Frame

Hull Frame

A second body-fixed frame $H = \{\mathbf{i}_H, \mathbf{j}_H, \mathbf{k}_H\}$ is attached to the internal hull. The origin of frame H is located at the hull center.

The angular velocity of the hull relative to the inertial frame is denoted as

$$\boldsymbol{\omega}_{H/O}.$$

Hull position relative to the inertial origin is given by

$$\mathbf{r}_{H/O}.$$

Sphere Frame

A body-fixed reference frame $S = \{\mathbf{i}_S, \mathbf{j}_S, \mathbf{k}_S\}$ is attached to the spherical shell. The origin of frame S is located at the sphere center. This frame rotates with the traction hoop and ribs assembly.

The angular velocity of the sphere relative to the inertial frame is denoted as

$$\boldsymbol{\omega}_{S/O}.$$

The sphere's position relative to the inertial origin is given by

$$\mathbf{r}_{S/O}.$$

System Center of Mass

Point C denotes the center of mass of the complete SSailOR system. Its position relative to the inertial origin is given by

$$\mathbf{r}_{C/O}.$$

The sphere's position relative to the Center of Mass is given by

$$\mathbf{r}_{S/C}.$$

Hull position relative to the Center of Mass is given by

$$\mathbf{r}_{H/C}.$$

Contact Points

Point P denotes the ground contact point of the sphere. Internal contact points BN and BP represent the mechanical interfaces between the sphere and the hull connectors. These points are used to define geometric constraints and to transmit internal forces between the two bodies.

The definition of these reference frames forms the basis for developing the kinematic relationships and applying Kane's method to derive the equations of motion.

5.4 Kinematics

The kinematic relations describe the translational and rotational motion of the SSailOR system independent of applied forces. The motion is defined through direction cosine matrices, position vectors, angular velocities, and their corresponding time derivatives.

5.4.1 Direction Cosine Matrices

The orientation of the hull frame H relative to the inertial frame O is defined using a sequence of rotations,

$$HCO = R_Y(\theta_H)\,R_X(\phi)\,R_Z(\psi) \tag{5.12}$$

The inverse transformation is

$$OCH = HCO^T \tag{5.13}$$

The orientation of the sphere frame S relative to the hull frame H is defined as

$$SCH = R_Y(\theta_S - \theta_H) \tag{5.14}$$

with inverse

$$HCS = SCH^T \tag{5.15}$$

The sphere orientation relative to the inertial frame is therefore

$$SCO = SCH.HCO \tag{5.16}$$

and

$$OCS = SCO^T \tag{5.17}$$

5.4.2 Position Vectors

The position vector of the sphere center relative to the inertial frame is

$$\mathbf{r}_{S/O} = x\mathbf{i}_O + y\mathbf{j}_O + z\mathbf{k}_O \tag{5.18}$$

where $x(t)$, $y(t)$, and $z(t)$ are time-dependent generalized coordinates.

Point C denotes the center of mass of the full SSailOR system. The position vector of C relative to the inertial frame is written as the sum

$$\mathbf{r}_{C/O} = \mathbf{r}_{C/S} + \mathbf{r}_{S/O} \tag{5.19}$$

where $\mathbf{r}_{C/S}$ is the position of C relative to the sphere center.

Expressed in the hull frame, the relative position vectors used in this work are

$$\mathbf{r}^H_{H/C} = \begin{bmatrix} 0 \\ 0 \\ -z_H \end{bmatrix}, \qquad \mathbf{r}^H_{S/C} = \begin{bmatrix} 0 \\ 0 \\ z_S \end{bmatrix} \tag{5.20}$$

and therefore

$$\mathbf{r}^H_{C/S} = -\mathbf{r}^H_{S/C}. \tag{5.21}$$

The hull center position relative to the inertial frame is then

$$\mathbf{r}_{H/O} = \mathbf{r}_{H/C} + \mathbf{r}_{C/O}. \tag{5.22}$$

5.4.3 Translational Velocities

The velocity of the sphere center in the inertial frame is

$$\bar{o}\mathbf{v}^{\mathbf{O}}{}_{S/O} = \frac{{}^{O}d}{dt}\mathbf{r}_{S/O} \tag{5.23}$$

Using the transport theorem,

$$\bar{o}\mathbf{v}^{\mathbf{S}}{}_{S/O} = \frac{{}^{S}d}{dt}\mathbf{r}_{S/O} + \boldsymbol{\omega}_{O/S} \times \mathbf{r}_{S/O} \tag{5.24}$$

The velocity of the hull center relative to the inertial frame is

$$\mathbf{v}_{H/O} = \mathbf{v}_{H/C} + \mathbf{v}_{C/O} \tag{5.25}$$

where

$$\bar{o}\mathbf{v}^{\mathbf{H}}{}_{C/O} = \frac{{}^{H}d}{dt}\mathbf{r}_{C/O} + \boldsymbol{\omega}_{O/H} \times \mathbf{r}_{C/O} \tag{5.26}$$

5.4.4 Translational Accelerations

The acceleration of the sphere center relative to the inertial frame is

$$\bar{o}\mathbf{a}^{\mathbf{O}}{}_{S/O} = \frac{{}^{O}d}{dt}\mathbf{v}_{S/O} \tag{5.27}$$

Applying the transport theorem,

$$\bar{o}\mathbf{a}^{\mathbf{S}}{}_{S/O} = \frac{{}^{S}d}{dt}\mathbf{v}^{\mathbf{S}}{}_{S/O} + \boldsymbol{\omega}_{O/S} \times \mathbf{v}^{\mathbf{S}}{}_{S/O} \tag{5.28}$$

Acceleration expressions for other points follow similarly from time differentiation of their velocity vectors.

5.4.5 Angular Velocity of Hull Relative to Inertial Frame

The angular velocity of the hull frame relative to the inertial frame is expressed in the hull frame as

$$\boldsymbol{\omega}_{O/H} = \omega_x \mathbf{i}_H + \omega_{yH} \mathbf{j}_H + \omega_z \mathbf{k}_H \tag{5.29}$$

The angular velocity components are extracted from the DCM using

$$\omega_x = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} HCOO\dot{C}H \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \tag{5.30}$$

$$\omega_{yH} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} HCOO\dot{C}H \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{5.31}$$

$$\omega_z = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} HCOO\dot{C}H \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{5.32}$$

5.4.6 Angular Acceleration

The angular acceleration of the hull relative to the inertial frame is

$$\boldsymbol{\alpha}_{O/H} = \frac{{}^{O}d}{dt} \boldsymbol{\omega}_{O/H} \tag{5.33}$$

5.4.7 Relative Motion of Sphere and Hull

The angular velocity of the sphere relative to the hull frame is

$$\boldsymbol{\omega}_{S/H} = \omega_{yS} \mathbf{j}_S \tag{5.34}$$

where

$$\omega_{yS} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} HCOO\dot{C}H \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{5.35}$$

The angular acceleration of the sphere relative to the inertial frame is obtained by time differentiation of the total angular velocity using the transport theorem.

5.5 Dynamic Model

5.5.1 Generalized Speeds and Coordinates

Generalized Coordinates

The generalized coordinates of the SSailOR system are defined as

$$q_1 = x = \text{Position of sphere center relative to } O \tag{5.36}$$
$$q_2 = y = \text{Position of sphere center relative to } O \tag{5.37}$$
$$q_3 = z = \text{Position of sphere center relative to } O \tag{5.38}$$
$$q_4 = \psi = \text{Yaw of hull } H \text{ relative to } O \tag{5.39}$$
$$q_5 = \phi = \text{Roll of hull } H \text{ relative to } O \tag{5.40}$$
$$q_6 = \theta_H = \text{Pitch of hull } H \text{ relative to } O \tag{5.41}$$
$$q_7 = \theta_S = \text{Pitch of sphere } S \text{ relative to } H \tag{5.42}$$

In vector form,

$$\mathbf{q} = \begin{bmatrix} x & y & z & \psi & \phi & \theta_H & \theta_S \end{bmatrix}^T \tag{5.43}$$

The generalized velocities are

$$\dot{\mathbf{q}} = \begin{bmatrix} \dot{x} & \dot{y} & \dot{z} & \dot{\psi} & \dot{\phi} & \dot{\theta}_H & \dot{\theta}_S \end{bmatrix}^T \tag{5.44}$$

Generalized Speeds

The generalized speeds are selected as

$$\mathbf{u} = \begin{bmatrix} u & v & w & \omega_z & \omega_x & \omega_{yH} & \omega_{yS} \end{bmatrix}^T \tag{5.45}$$

Explicitly,

$$u_1 = u \tag{5.46}$$

$$u_2 = v \tag{5.47}$$

$$u_3 = w \tag{5.48}$$

$$u_4 = \omega_z \tag{5.49}$$

$$u_5 = \omega_x \tag{5.50}$$

$$u_6 = \omega_{yH} \tag{5.51}$$

$$u_7 = \omega_{yS} \tag{5.52}$$

The translational velocity of the sphere center expressed in the inertial frame is

$$\mathbf{v}_{S/O}^{O} = \dot{x}\mathbf{i}_O + \dot{y}\mathbf{j}_O + \dot{z}\mathbf{k}_O \tag{5.53}$$

Using the direction cosine matrix HCO, the velocity expressed in the hull frame becomes

$$\mathbf{v}_{S/O}^{H} = HCO.\mathbf{v}_{S/O}^{O} = u\mathbf{i}_H + v\mathbf{j}_H + w\mathbf{k}_H \tag{5.54}$$

The velocity components are therefore:

$$u = \dot{x}(\cos\psi\cos\theta_H - \sin\phi\sin\psi\sin\theta_H) + \dot{y}(\cos\theta_H\sin\psi + \cos\psi\sin\phi\sin\theta_H) - \dot{z}\cos\phi\sin\theta_H \tag{5.55}$$

$$v = \dot{z}\sin\phi + \dot{y}\cos\phi\cos\psi - \dot{x}\cos\phi\sin\psi \tag{5.56}$$

$$w = \dot{x}(\cos\psi\sin\theta_H + \cos\theta_H\sin\phi\sin\psi) + \dot{y}(\sin\psi\sin\theta_H - \cos\psi\cos\theta_H\sin\phi) + \dot{z}\cos\phi\cos\theta_H \tag{5.57}$$

The angular velocity of the hull relative to the inertial frame expressed in the hull frame is

$$\boldsymbol{\omega}_{H/O} = \omega_x\mathbf{i}_H + \omega_{yH}\mathbf{j}_H + \omega_z\mathbf{k}_H \tag{5.58}$$

Therefore, the components are:

$$\omega_x = \dot{\phi}\cos\theta_H - \dot{\psi}\cos\phi\sin\theta_H \tag{5.59}$$

$$\omega_{yH} = \dot{\theta}_H + \dot{\psi}\sin\phi \tag{5.60}$$

$$\omega_z = \dot{\phi}\sin\theta_H + \dot{\psi}\cos\phi\cos\theta_H \tag{5.61}$$

The angular velocity of the sphere relative to the hull frame is

$$\boldsymbol{\omega}_{S/H} = \omega_{yS}\mathbf{j}_H \tag{5.62}$$

where

$$\omega_{yS} = \dot{\theta}_S - \dot{\theta}_H \tag{5.63}$$

Kane's method relates generalized speeds to generalized velocities through

$$[\mathbf{u}] =]\mathbf{Y}][\dot{\mathbf{q}}] - [\mathbf{Z}] \tag{5.64}$$

For the present system,

$$\mathbf{Z} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T \tag{5.65}$$

The matrix $\mathbf{Y}$ is a 7×7 matrix defined as

$$\mathbf{Y} = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} & 0 & 0 & 0 & 0 \\ Y_{21} & Y_{22} & Y_{23} & 0 & 0 & 0 & 0 \\ Y_{31} & Y_{32} & Y_{33} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Y_{44} & Y_{45} & 0 & 0 \\ 0 & 0 & 0 & Y_{54} & Y_{55} & 0 & 0 \\ 0 & 0 & 0 & 0 & Y_{65} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix} \tag{5.66}$$

$$Y_{11} = \cos q_4 \cos q_6 - \sin q_5 \sin q_4 \sin q_6 \tag{5.67}$$

$$Y_{12} = \cos q_6 \sin q_4 + \cos q_4 \sin q_5 \sin q_6 \tag{5.68}$$

$$Y_{13} = -\cos q_5 \sin q_6 \tag{5.69}$$

$$Y_{21} = -\cos q_5 \sin q_4 \tag{5.70}$$

$$Y_{22} = \cos q_5 \cos q_4 \tag{5.71}$$

$$Y_{23} = \sin q_5 \tag{5.72}$$

$$\tag{5.73}$$

$$Y_{31} = \cos q_4 \sin q_6 + \cos q_6 \sin q_5 \sin q_4 \tag{5.74}$$

$$Y_{32} = \sin q_4 \sin q_6 - \cos q_4 \cos q_6 \sin q_5 \tag{5.75}$$

$$Y_{33} = \cos q_5 \cos q_6 \tag{5.76}$$

$$Y_{44} = \sin q_6 \tag{5.77}$$

$$Y_{45} = \cos q_5 \cos q_6 \tag{5.78}$$

$$Y_{54} = \cos q_6 \tag{5.79}$$

$$Y_{55} = -\cos q_5 \sin q_6 \tag{5.80}$$

$$Y_{65} = \sin q_5 \tag{5.81}$$

The generalized velocities are recovered from

$$[\dot{\mathbf{q}}] = [\mathbf{W}][\mathbf{u}] \tag{5.82}$$

$$[\mathbf{W}] = [\mathbf{Y}]^{-1} \tag{5.83}$$

$$\mathbf{W} = \begin{bmatrix} W_{11} & W_{12} & W_{13} & 0 & 0 & 0 & 0 \\ W_{21} & W_{22} & W_{23} & 0 & 0 & 0 & 0 \\ W_{31} & W_{32} & W_{33} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & W_{44} & W_{45} & 0 & 0 \\ 0 & 0 & 0 & W_{54} & W_{55} & 0 & 0 \\ 0 & 0 & 0 & W_{64} & W_{65} & 1 & 0 \\ 0 & 0 & 0 & W_{74} & W_{75} & 1 & -1 \end{bmatrix} \tag{5.84}$$

$$W_{11} = \cos q_4 \cos q_6 - \sin q_5 \sin q_4 \sin q_6 \tag{5.85}$$

$$W_{12} = -\cos q_5 \sin q_4 \tag{5.86}$$

$$W_{13} = \cos q_4 \sin q_6 + \cos q_6 \sin q_5 \sin q_4 \tag{5.87}$$

$$W_{21} = \cos q_6 \sin q_4 + \cos q_4 \sin q_5 \sin q_6 \tag{5.88}$$

$$W_{22} = \cos q_5 \cos q_4 \tag{5.89}$$

$$W_{23} = \sin q_4 \sin q_6 - \cos q_4 \cos q_6 \sin q_5 \tag{5.90}$$

$$\tag{5.91}$$

$$W_{31} = -\cos q_5 \sin q_6 \tag{5.92}$$

$$W_{32} = \sin q_5 \tag{5.93}$$

$$W_{33} = \cos q_5 \cos q_6 \tag{5.94}$$

$$W_{44} = \sin q_6 \tag{5.95}$$

$$W_{45} = \cos q_6 \tag{5.96}$$

$$W_{54} = \frac{\cos q_6}{\cos q_5} \tag{5.97}$$

$$W_{55} = -\frac{\sin q_6}{\cos q_5} \tag{5.98}$$

$$W_{64} = -\frac{\cos q_6 \sin q_5}{\cos q_5} \tag{5.99}$$

$$W_{65} = \frac{\sin q_5 \sin q_6}{\cos q_5} \tag{5.100}$$

$$W_{74} = -\frac{\cos q_6 \sin q_5}{\cos q_5} \tag{5.101}$$

$$W_{75} = \frac{\sin q_5 \sin q_6}{\cos q_5} \tag{5.102}$$

These 7 generalized speeds result in 7 governing Kane's method equations $Fr + Fr^* = 0$, one for each of the generalized speeds, u1, ...u7.

$$F_1 + F_1^* = 0 \tag{5.103}$$

$$F_2 + F_2^* = 0 \tag{5.104}$$

$$F_3 + F_3^* = 0 \tag{5.105}$$

$$F_4 + F_4^* = 0 \tag{5.106}$$

$$F_5 + F_5^* = 0 \tag{5.107}$$

$$F_6 + F_6^* = 0 \tag{5.108}$$

$$F_7 + F_7^* = 0 \tag{5.109}$$

5.5.2 Partial Velocities and Partial Angular Velocities

To apply Kane's method, the translational velocities, angular velocities, accelerations, and angular accelerations are expressed in terms of the generalized coordinates $(q_1, \ldots, q_7)$, generalized speeds $(u_1, \ldots, u_7)$, and their time derivatives $(\dot{u}_1, \ldots, \dot{u}_7)$.

For each rigid body in the system, the velocity of its center of mass and its angular velocity are

written as linear combinations of the generalized speeds. The coefficients of these generalized speeds are defined as the partial velocities and partial angular velocities.

Rigid Body Partial Velocities

For a rigid body k, the velocity of its center of mass relative to the inertial frame is written as

$$\mathbf{v}_{cm_k/O} = \sum_{r=1}^{7} \mathbf{v}_{cm_k,r}\, u_r + \mathbf{v}_{cm_k,t} \tag{5.110}$$

The rth partial velocity is defined as

$$\bar{o}_o \mathbf{v}_{cm_k,r} = \frac{\partial}{\partial u_r}\left(\bar{o}\mathbf{v}_{cm_k/O}\right) \tag{5.111}$$

Hull Center of Mass

For the hull body, whose center of mass is located at point H,

$$\bar{o}_o \mathbf{v}_{H/O,r} = \frac{\partial}{\partial u_r}\left(\bar{o}\mathbf{v}_{H/O}\right), \qquad r = 1,\ldots,7 \tag{5.112}$$

Explicitly,

$$\bar{o}_o \mathbf{v}_{H/O,1} = \frac{\partial\, \bar{o}\mathbf{v}_{H/O}}{\partial u_1} \tag{5.113}$$

$$\bar{o}_o \mathbf{v}_{H/O,2} = \frac{\partial\, \bar{o}\mathbf{v}_{H/O}}{\partial u_2} \tag{5.114}$$

$$\vdots \tag{5.115}$$

$$\bar{o}_o \mathbf{v}_{H/O,7} = \frac{\partial\, \bar{o}\mathbf{v}_{H/O}}{\partial u_7} \tag{5.116}$$

Sphere Center of Mass

For the sphere body, whose center of mass is located at point S,

$$\bar{o}_o \mathbf{v}_{S/O,r} = \frac{\partial}{\partial u_r}\left(\bar{o}\mathbf{v}_{S/O}\right), \qquad r = 1,\ldots,7 \tag{5.117}$$

Similarly,

$$\bar{o}_o \mathbf{v}_{S/O,1} = \frac{\partial\, \bar{o}\mathbf{v}_{S/O}}{\partial\, u_1} \tag{5.118}$$

$$\bar{o}_o \mathbf{v}_{S/O,2} = \frac{\partial\, \bar{o}\mathbf{v}_{S/O}}{\partial\, u_2} \tag{5.119}$$

$$\vdots \tag{5.120}$$

$$\bar{o}_o \mathbf{v}_{S/O,7} = \frac{\partial\, \bar{o}\mathbf{v}_{S/O}}{\partial\, u_7} \tag{5.121}$$

Rigid Body Partial Angular Velocities

For a rigid body k, the angular velocity relative to the inertial frame is written as

$$\boldsymbol{\omega}_{k/O} = \sum_{r=1}^{7} \boldsymbol{\omega}_{k,r}\, u_r + \boldsymbol{\omega}_{k,t} \tag{5.122}$$

The rth partial angular velocity is defined as

$$\boldsymbol{\omega}_{k,r} = \frac{\partial}{\partial\, u_r}\left(\boldsymbol{\omega}_{k/O}\right) \tag{5.123}$$

Hull Angular Partial Velocities

For the hull,

$$\boldsymbol{\omega}_{O/H,r} = \frac{\partial}{\partial\, u_r}\left(\boldsymbol{\omega}_{O/H}\right), \qquad r = 1, \ldots, 7 \tag{5.124}$$

Explicitly,

$$\omega_{O/H,1} = \frac{\partial\, \omega_{O/H}}{\partial\, u_1} \tag{5.125}$$

$$\omega_{o/H,2} = \frac{\partial\, \omega_{O/H}}{\partial\, u_2} \tag{5.126}$$

$$\vdots \tag{5.127}$$

$$\omega_{O/H,7} = \frac{\partial\, \omega_{O/H}}{\partial\, u_7} \tag{5.128}$$

Sphere Angular Partial Velocities

For the sphere,

$$\boldsymbol{\omega}_{O/S,r} = \frac{\partial}{\partial u_r}\left(\boldsymbol{\omega}_{O/S}\right), \qquad r = 1, \ldots, 7 \tag{5.129}$$

Explicitly,

$$\omega_{O/S,1} = \frac{\partial \omega_{O/S}}{\partial u_1} \tag{5.130}$$

$$\omega_{o/S,2} = \frac{\partial \omega_{O/S}}{\partial u_2} \tag{5.131}$$

$$\vdots \tag{5.132}$$

$$\omega_{O/S,7} = \frac{\partial \omega_{O/S}}{\partial u_7} \tag{5.133}$$

5.5.3 Forces acting on Hull

The hull body of SSailOR is subjected to aerodynamic, gravitational, and internal interaction forces. These forces are required for the formulation of the generalized active forces in Kane's method.

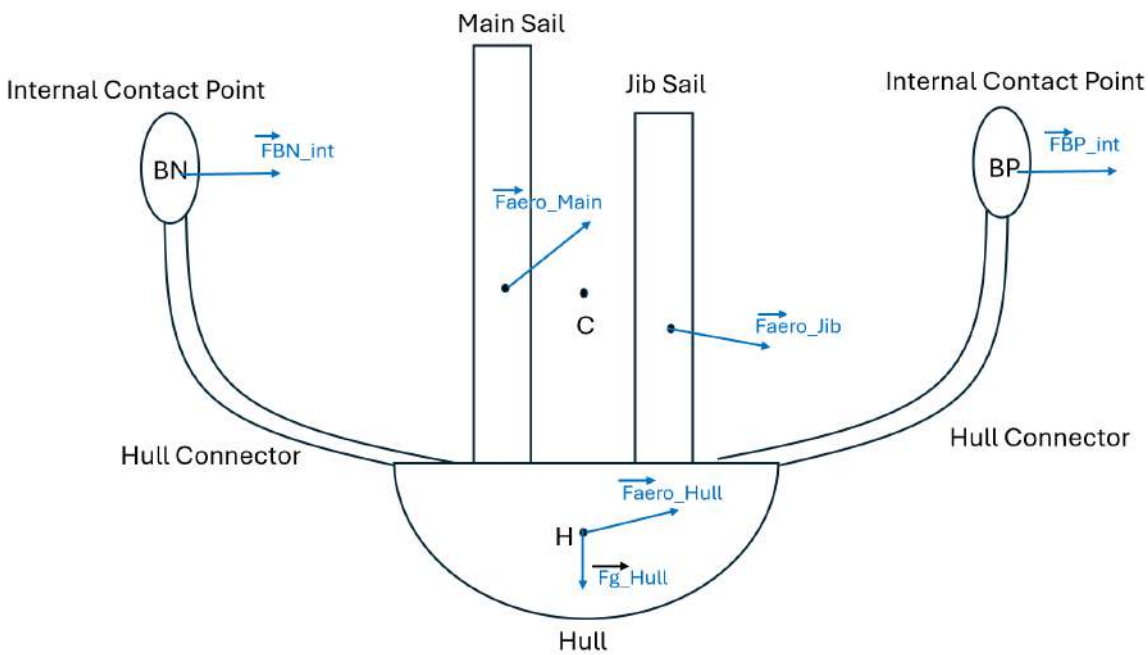


Figure 5.3: Hull Force Illustration

The forces acting on the hull, as shown in Fig. 5.3, are:

- Aerodynamic force on the hull structure: $\mathbf{F}_{\text{aero,hull}}$
- Aerodynamic force on the main sail: $\mathbf{F}_{\text{aero,main}}$
- Aerodynamic force on the jib sail: $\mathbf{F}_{\text{aero,jib}}$
- Internal reaction force at point BP: $\mathbf{F}_{BP,\text{int}}$
- Internal reaction force at point BN: $\mathbf{F}_{BN,\text{int}}$
- Gravitational force on the hull: $\mathbf{F}_{g,\text{hull}}$

Gravitational Force on Hull

The gravitational force acts at the center of mass of the hull and is expressed in the inertial frame as

$$\mathbf{F}_{g,\text{hull}} = m_H(-g)\mathbf{k}_O \tag{5.134}$$

where m_H is the mass of the hull and g is gravitational acceleration.

Apparent Wind Velocity

The aerodynamic forces depend on the apparent wind velocity at the sail surfaces.

The orientation matrix between the hull frame H and the inertial frame O is

$$HCO = R_Y(\theta_H)R_X(\phi)R_Z(\psi) \tag{5.135}$$

with inverse

$$OCH = HCO^T \tag{5.136}$$

The free-stream wind velocity in the inertial frame is defined as

$$\mathbf{V}_\infty^O = \begin{bmatrix} 0 \\ -V_\infty \\ 0 \end{bmatrix} \tag{5.137}$$

Transforming into the hull frame,

$$\mathbf{V}_\infty^H = HCO.\mathbf{V}_\infty^O \tag{5.138}$$

The apparent wind at the sail location is

$$\mathbf{V}_{\text{app,sails}} = \mathbf{V}_{\infty}^{H} - \mathbf{v}_{S/O}^{H} \tag{5.139}$$

where $\mathbf{v}_{S/O}^{H}$ is the velocity of the sphere center expressed in the hull frame.
The magnitude of the apparent velocity is

$$V_{\text{mag}} = \|\mathbf{V}_{\text{app,sails}}\| \tag{5.140}$$

Lift and Drag Directions for Sails

The drag direction is defined as

$$\mathbf{e}_D = \frac{\mathbf{V}_{\text{app,sails}}}{\|\mathbf{V}_{\text{app,sails}}\|} \tag{5.141}$$

Assuming the sail span direction is

$$\mathbf{e}_{span} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{5.142}$$

The lift direction is obtained from

$$\mathbf{e}_L = \frac{\mathbf{e}_{span} \times \mathbf{e}_D}{\|\mathbf{e}_{span} \times \mathbf{e}_D\|} \tag{5.143}$$

Aerodynamic Forces on the Sails

The aerodynamic forces acting on the main and jib sails are computed using lift and drag coefficients identified from Wind Tunnel Test 2 results. The lift coefficients are modeled as linear functions of the sail angle of attack (in degrees):

$$C_{L,\text{main}} = 0.047\,\alpha_{m,\text{deg}} + 0.063 \tag{5.144}$$

$$C_{L,\text{jib}} = 0.019\,\alpha_{j,\text{deg}} + 0.091 \tag{5.145}$$

The drag coefficients are modeled using a finite-wing induced drag formulation:

$$C_{D,\text{main}} = 0.06 + \frac{C_{L,\text{main}}^2}{\pi AR_{\text{main}} e} \tag{5.146}$$

$$C_{D,\text{jib}} = 0.06 + \frac{C_{L,\text{jib}}^2}{\pi AR_{\text{jib}} e} \tag{5.147}$$

where

$$AR_{\text{main}} = 2.1, \qquad AR_{\text{jib}} = 2.41, \qquad e = 0.69$$

The aerodynamic force on the main sail is then computed as

$$\mathbf{F}_{\text{aero,main}} = \frac{1}{2}\rho V_{\text{mag}}^2 C_{L,\text{main}} S_{\text{main}} \mathbf{e}_L + \frac{1}{2}\rho V_{\text{mag}}^2 C_{D,\text{main}} S_{\text{main}} \mathbf{e}_D \tag{5.148}$$

where

$$S_{\text{main}} = 0.03444 \text{ m}^2$$

Similarly, the aerodynamic force on the jib sail is

$$\mathbf{F}_{\text{aero,jib}} = \frac{1}{2}\rho V_{\text{mag}}^2 C_{L,\text{jib}} S_{\text{jib}} \mathbf{e}_L + \frac{1}{2}\rho V_{\text{mag}}^2 C_{D,\text{jib}} S_{\text{jib}} \mathbf{e}_D \tag{5.149}$$

with

$$S_{\text{jib}} = 0.01903 \text{ m}^2$$

and air density

$$\rho = 1.15 \text{ kg/m}^3.$$

Here, $\mathbf{e}_L$ and $\mathbf{e}_D$ denote the unit lift and drag directions determined from the apparent wind vector.

Internal Interaction Forces for Hull

The internal interaction forces for the hull is defined in the hull frame as

$$\mathbf{F}_{BP,\text{int,hull}} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \mathbf{F}_{BN,\text{int,hull}} = \begin{bmatrix} f_4 \\ 0 \\ f_6 \end{bmatrix} \tag{5.150}$$

Aerodynamic Force on Hull Structure

The aerodynamic drag acting directly on the hull structure is modeled using a vectorized drag formulation.

The apparent wind at the Hull location is

$$\mathbf{V}_{\text{app,hull}} = \mathbf{V}_{\infty}^{H} - \mathbf{v}_{H/O}^{H} \tag{5.151}$$

The dynamic pressure term is

$$Q_{\text{hull}} = \frac{1}{2}\rho\, V_{\text{app,hull}}^{2} \tag{5.152}$$

The aerodynamic force on the hull is

$$\mathbf{F}_{\text{aero,hull}} = Q_{\text{hull}} \begin{bmatrix} u_{\text{hull}} S_i C_{D,i} \\ v_{\text{hull}} S_j C_{D,j} \\ w_{\text{hull}} S_k C_{D,k} \end{bmatrix} \tag{5.153}$$

The drag coefficients $C_{D,i}$ and $C_{D,j}$ were identified using coefficient identification techniques based on the wind tunnel Test 2 results and Varanwal et al. (2026)

5.5.4 Forces on Sphere

The sphere body of SSailOR is subjected to aerodynamic, gravitational,internal interaction, and ground contact forces. These forces are required for the formulation of the generalized active forces in Kane's method.

The forces acting on the hull, as shown in Fig. 5.4, are:

- Aerodynamic force on the sphere: $\mathbf{F}_{\text{aero,sphere}}$
- Internal reaction force at point BP: $\mathbf{F}_{BP,\text{int}}$
- Internal reaction force at point BN: $\mathbf{F}_{BN,\text{int}}$
- Gravitational force on the sphere: $\mathbf{F}_{g,\text{sphere}}$
- Ground contact force on the sphere: $\mathbf{F}_{p,\text{sphere}}$

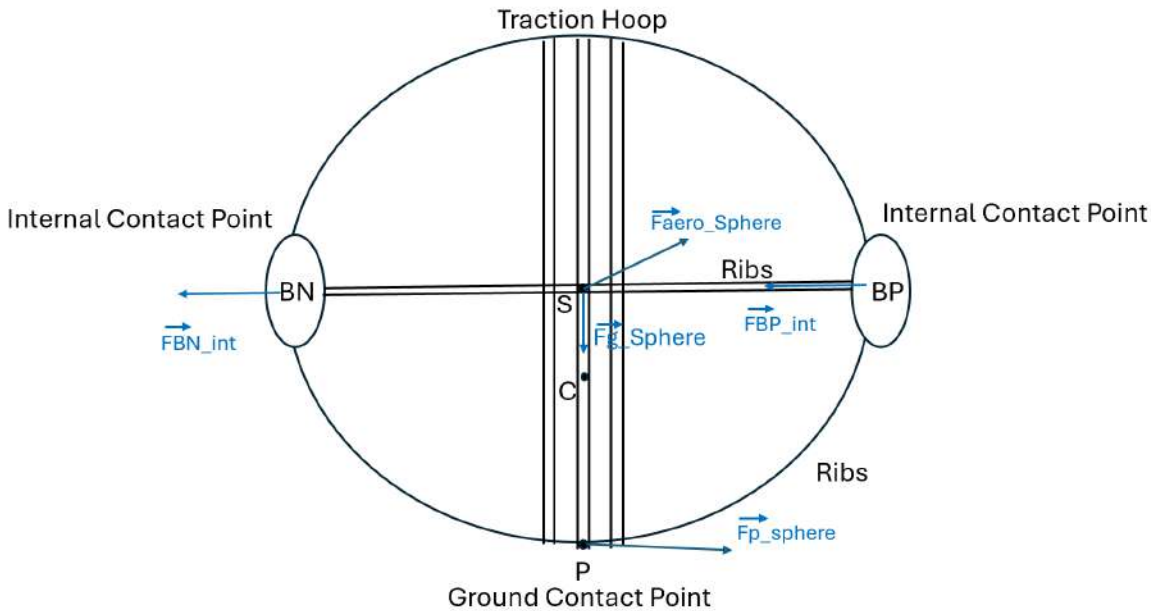


Figure 5.4: Sphere Force Illustration

Aerodynamic Forces on Sphere Structure

The total aerodynamic force on the sphere is composed of contributions from the ribs, the traction hoop, and side plates.

$$\mathbf{F}_{\text{aero,sphere}} = \mathbf{F}_{\text{aero,rib}} + \mathbf{F}_{\text{aero,hoop}} + \mathbf{F}_{\text{aero,plate}} \tag{5.154}$$

Each component is modeled using a vectorized drag formulation similar to that used for the hull. The drag coefficients for the ribs, hoop, and plates were identified using coefficient identification methods based on wind tunnel Test 2 results and Varanwal et al. (2026).

The aerodynamic force on the plates is

$$\mathbf{F}_{\text{aero,plate}} = Q_{\text{plate}} \begin{bmatrix} u_{\text{plate}} S_{i,\text{plate}} C_{D,i,\text{plate}} \\ v_{\text{plate}} S_{j,\text{plate}} C_{D,j,\text{plate}} \\ w_{\text{plate}} S_{k,\text{plate}} C_{D,k,\text{plate}} \end{bmatrix} \tag{5.155}$$

The aerodynamic force on the Hoop is

$$\mathbf{F}_{\text{aero,hoop}} = Q_{\text{hoop}} \begin{bmatrix} u_{\text{hoop}} S_{i,\text{hoop}} C_{D,i,\text{hoop}} \\ v_{\text{hoop}} S_{j,\text{hoop}} C_{D,j,\text{hoop}} \\ w_{\text{hoop}} S_{k,\text{hoop}} C_{D,k,\text{hoop}} \end{bmatrix} \tag{5.156}$$

The aerodynamic force on the rib is

$$\mathbf{F}_{\text{aero,rib}} = Q_{\text{rib}} \begin{bmatrix} u_{\text{rib}} S_{i,\text{rib}} C_{D,i,\text{rib}} \\ v_{\text{rib}} S_{j,\text{rib}} C_{D,j,\text{rib}} \\ w_{\text{rib}} S_{k,\text{rib}} C_{D,k,\text{rib}} \end{bmatrix} \tag{5.157}$$

Gravitational Force

The gravitational force on the sphere acts at its center of mass and is expressed in the inertial frame as

$$\mathbf{F}_{g,\text{sphere}} = m_S(-g)\mathbf{k}_O \tag{5.158}$$

where m_S is the mass of the sphere.

Internal Interaction Forces

The internal interaction forces between the sphere and hull satisfy Newton's third law. The sphere's internal force is equal and opposite to that of the hull's internal force.

$$\mathbf{F}_{BP,\text{int,sphere}} = \begin{bmatrix} -f_1 \\ -f_2 \\ -f_3 \end{bmatrix} \mathbf{F}_{BN,\text{int,sphere}} = \begin{bmatrix} -f_4 \\ 0 \\ -f_6 \end{bmatrix} \tag{5.159}$$

Ground Contact Force

The ground contact force at point P is expressed in the inertial frame as

$$\mathbf{F}_P = \begin{bmatrix} F_{\text{fr},x} \\ F_{\text{fr},y} \\ F_n \end{bmatrix} \tag{5.160}$$

where $F_{\text{fr},x}$ and $F_{\text{fr},y}$ are friction forces and F_n is the normal reaction force.

5.5.5 Torques on Hull and Sphere

The torques of the hull and sphere require calculation of moments about their respective centers of mass. All moment arms are expressed in the hull frame unless otherwise specified.

Torque Acting on the Hull

The position vectors required for moment computation are defined as follows.

The distance from the main sail quarter chord point to the sphere CM is

$$\mathbf{r}^H_{M/S} = \begin{bmatrix} -d\,m \\ 0 \\ 0 \end{bmatrix} \tag{5.161}$$

The distance from the jib sail quarter chord point to the sphere CM is

$$\mathbf{r}^H_{J/S} = \begin{bmatrix} d\,j \\ 0 \\ 0 \end{bmatrix} \tag{5.162}$$

The position vectors of the sail quarter chord relative to the hull center are therefore

$$\mathbf{r}^H_{M/H} = \mathbf{r}^H_{M/S} + \mathbf{r}^H_{S/H} \tag{5.163}$$

$$\mathbf{r}^H_{J/H} = \mathbf{r}^H_{J/S} + \mathbf{r}^H_{S/H} \tag{5.164}$$

The sphere–hull internal contact points relative to the sphere center are

$$\mathbf{r}^H_{BP/S} = \begin{bmatrix} 0 \\ R \\ 0 \end{bmatrix} \qquad \mathbf{r}^H_{BN/S} = \begin{bmatrix} 0 \\ -R \\ 0 \end{bmatrix} \tag{5.165}$$

The sphere center relative to the hull center is

$$\mathbf{r}^H_{S/H} = \begin{bmatrix} 0 \\ 0 \\ s_h \end{bmatrix} \tag{5.166}$$

Thus, the contact points relative to the hull center are

$$\mathbf{r}^H_{BP/H} = \mathbf{r}^H_{BP/S} + \mathbf{r}^H_{S/H} \tag{5.167}$$

$$\mathbf{r}^H_{BN/H} = \mathbf{r}^H_{BN/S} + \mathbf{r}^H_{S/H} \tag{5.168}$$

The total torque acting on the hull about its center of mass is

$$\mathbf{T}_{\text{hull}} = \mathbf{r}^H_{M/H} \times \mathbf{F}_{\text{aero,main}} + \mathbf{r}^H_{J/H} \times \mathbf{F}_{\text{aero,jib}} + \mathbf{r}^H_{BP/H} \times \mathbf{F}_{BP,\text{int,hull}} + \mathbf{r}^H_{BN/H} \times \mathbf{F}_{BN,\text{int,hull}} \tag{5.169}$$

The gravitational force is not included in the moment calculation since it acts at the hull center of mass and therefore produces no moment about that point.

Torque Acting on the Sphere

For the sphere, moments are computed about the sphere's center of mass.

The ground contact point relative to the sphere center in the inertial frame is

$$\mathbf{r}^O_{P/S} = \begin{bmatrix} 0 \\ 0 \\ -R \end{bmatrix} \tag{5.170}$$

$$\mathbf{r}^H_{P/S} = HCO.\mathbf{r}^O_{P/S} \tag{5.171}$$

The total torque acting on the sphere about its center of mass is

$$\mathbf{T}_{\text{sphere}} = \mathbf{r}^H_{P/S} \times \mathbf{F}_P + \mathbf{r}^H_{BP/S} \times \mathbf{F}_{BP,\text{int,sphere}} + \mathbf{r}^H_{BN/S} \times \mathbf{F}_{BN,\text{int,sphere}} \tag{5.172}$$

The gravitational force on the sphere is excluded from the moment equation because it acts through the center of mass. The aerodynamic drag forces on the sphere structure are also neglected in the torque calculation since they act approximately at or very near the sphere's center of mass and therefore contribute a negligible moment about that point.

5.5.6 Solving For Internal Contact Forces

To solve for the unknown forces $f1$, $f2$, $f3$, $f4$, and $f6$, the Newton–Euler equations of motion are applied to the hull body.

Translational Equations of Motion

The translational dynamics of the hull in the hull frame are written as

$$\mathbf{F}_{\text{hull}} = m_H.\bar{o}\mathbf{a}^H_{H/O} \tag{5.173}$$

Rearranging,

$$\mathbf{F}_{\text{hull}} - m_H.\bar{o}\mathbf{a}^H_{H/O} = \mathbf{0} \tag{5.174}$$

Expanding component-wise yields three scalar equations:

$$\text{eq}_1 = 0 \tag{5.175}$$
$$\text{eq}_2 = 0 \tag{5.176}$$
$$\text{eq}_3 = 0 \tag{5.177}$$

Rotational Equations of Motion

The rotational dynamics of the hull about its center of mass are written as

$$\mathbf{T}_{\text{hull}} = \mathbf{I}_H.\boldsymbol{\alpha}_{O/H} \tag{5.178}$$

Rearranging,

$$\mathbf{T}_{\text{hull}} - \mathbf{I}_H.\boldsymbol{\alpha}_{O/H} = \mathbf{0} \tag{5.179}$$

For the present system, only the x and z components of the torque equation are required to close the system of equations, yielding two additional scalar equations:

$$\text{eq}_4 = 0 \tag{5.180}$$
$$\text{eq}_5 = 0 \tag{5.181}$$

Solution Procedure

The five scalar equations

$$\text{eq}_1, \text{eq}_2, \text{eq}_3, \text{eq}_4, \text{eq}_5$$

are solved simultaneously for the five unknown internal force components:

$$f_1,\ f_2,\ f_3,\ f_4,\ f_6$$

This yields closed-form expressions of the internal contact forces in terms of the generalized coordinates, generalized speeds, and their time derivatives.

Symbolically,

$$\{f_1, f_2, f_3, f_4, f_6\} = f(\mathbf{q}, \mathbf{u}, \dot{\mathbf{u}}) \tag{5.182}$$

These expressions are substituted into the generalized active force formulation, eliminating the internal forces from the final Kane equations of motion.

5.5.7 Solving for Ground Contact Forces

The ground contact force at point P is modeled as

$$\mathbf{F}_P^O = \begin{bmatrix} F_{\text{fr},x} \\ F_{\text{fr},y} \\ F_n \end{bmatrix} \tag{5.183}$$

where $F_{\text{fr},x}$ and $F_{\text{fr},y}$ are the friction components in the horizontal directions and F_n is the normal reaction.

This force is expressed in the H frame :

$$\mathbf{F}_P^H = HCO. \begin{bmatrix} F_{\text{fr},x} \\ F_{\text{fr},y} \\ F_n \end{bmatrix} \tag{5.184}$$

To determine these three unknowns, the translational equation of motion is written for the complete SSailOR system (sphere + hull) using the system center of mass C.

Total External Force

The total external force acting on the system, expressed in the hull frame, is written as

$$\mathbf{F}_C^H = \mathbf{F}_{\text{aero,main}}^H + \mathbf{F}_{\text{aero,jib}}^H + \mathbf{F}_{\text{aero,hull}}^H + \mathbf{F}_{\text{aero,sphere}}^H + \mathbf{F}_{g,\text{hull}}^H + \mathbf{F}_{g,\text{sphere}}^H + \mathbf{F}_P^H \tag{5.185}$$

where $\mathbf{F}_{g,\text{hull}}^H$ and $\mathbf{F}_{g,\text{sphere}}^H$ are the gravitational forces expressed in the hull frame.

Acceleration of the System Center of Mass

The acceleration of the system center of mass C relative to the inertial frame is computed using the transport theorem, since $\mathbf{v}_{C/O}$ is expressed in the rotating hull frame H:

$$\bar{o}\mathbf{a}^H_{C/O} = \frac{{}^H d}{d\,t}\left(\mathbf{v}^H_{C/O}\right) + \boldsymbol{\omega}^H_{O/H} \times \mathbf{v}^H_{C/O} \tag{5.186}$$

Force Balance and Solution

Applying Newton's second law to the full system gives

$$\mathbf{F}_C = m_T \mathbf{a}^H_{C/O} \tag{5.187}$$

where $m_T = m_H + m_S$ is the total system mass.

Rearranging,

$$\mathbf{F}_C - m_T \mathbf{a}^H_{C/O} = \mathbf{0} \tag{5.188}$$

Expanding component wise produces three scalar equations,

$$\mathrm{eq}_1 = 0 \tag{5.189}$$

$$\mathrm{eq}_2 = 0 \tag{5.190}$$

$$\mathrm{eq}_3 = 0 \tag{5.191}$$

which are solved simultaneously for the three unknown ground contact force components:

$$F_{\mathrm{fr},x}, \quad F_{\mathrm{fr},y}, \quad F_n$$

Symbolically, the solution may be written as

$$\{F_{\mathrm{fr},x}, F_{\mathrm{fr},y}, F_n\} = f(\mathbf{q}, \mathbf{u}, \dot{\mathbf{u}}) \tag{5.192}$$

The resulting expressions are substituted into the final equations of motion, eliminating the ground contact force unknowns from the governing Kane equations.

5.5.8 No Contact Break Constraint

To ensure the sphere does not lose contact with the ground, an acceleration-level contact constraint is enforced at the ground contact point P. Following the constraint formulation described in (https://physics.stackexchange.com/users/392/john alexiou), the vertical spatial

acceleration of the point on the sphere that is instantaneously in contact with the ground must be zero while contact is maintained.

Let the position vector from the sphere center S to the ground contact point be expressed in the inertial frame as

$$\mathbf{r}^{O}_{P/S} = \begin{bmatrix} 0 \\ 0 \\ -R \end{bmatrix}. \tag{5.193}$$

The translational acceleration of the sphere center is available in the hull frame, $\bar{o}\mathbf{a}^{H}_{S/O}$, and is transformed to the inertial frame using

$$\bar{o}\mathbf{a}^{O}_{S/O} = OCH\,\mathbf{a}^{H}_{S/O}. \tag{5.194}$$

Similarly, the angular acceleration of the sphere is transformed to the inertial frame as

$$\boldsymbol{\alpha}^{O}_{S/O} = OCH\,\boldsymbol{\alpha}^{H}_{S/O}. \tag{5.195}$$

Neglecting higher-order terms as implemented in the model, the spatial acceleration of the contact point due to sphere translation and angular acceleration is written as

$$\bar{o}\mathbf{a}^{O}_{P*} = \bar{o}\mathbf{a}^{O}_{S/O} + \boldsymbol{\alpha}^{O}_{S/O} \times \mathbf{r}^{O}_{P/S}. \tag{5.196}$$

The no-ground-break condition is enforced by constraining the vertical component of this acceleration to zero:

$$\Phi(\mathbf{q}, \mathbf{u}, \dot{\mathbf{u}}) \equiv \left(\bar{o}\mathbf{a}^{O}_{P*}\right).\mathbf{k}_o = 0. \tag{5.197}$$

Equation (5.197), together with the seven Kane equations

$$F_r + F^*_r = 0, \qquad r = 1, \ldots, 7,$$

forms an augmented set of equations used to solve for the generalized speed derivatives $\dot{\mathbf{u}}$.

5.6 Simulation

5.6.1 Simulation Setup

Model Parameters

Table 5.1 summarizes the parameters used in the dynamic simulation. The rover is modeled as a two-body system consisting of a hull and a spherical body. The geometric and inertial parameters were selected based on the physical SSailOR 0.4 scaled prototype.

Table 5.1: Simulation Parameters

Symbol	Value	Description
R	0.2	Sphere radius (m)
s_h	0.1070	Distance from sphere CM to hull CM along $\mathbf{k}_H$ (m)
m_H	4	Hull mass (kg)
m_S	2	Sphere mass (kg)
d_m	0.034	$\mathbf{i}_H$-offset from sphere CM to main-sail quarter-chord (m)
d_j	0.1272	$\mathbf{i}_H$-offset from sphere CM to jib-sail quarter-chord (m)
z_S	0.07133	Distance from system CM to sphere CM along $\mathbf{k}_H$ (m)
z_H	0.03567	Distance from system CM to hull CM along $\mathbf{k}_H$ (m)
$I_{xx,H}$	0.043	Hull principal inertia about $\mathbf{i}_H$ (kg·m^2)
$I_{yy,H}$	0.038	Hull principal inertia about $\mathbf{j}_H$ (kg·m^2)
$I_{zz,H}$	0.036	Hull principal inertia about $\mathbf{k}_H$ (kg·m^2)
$I_{xx,S}$	0.02	Sphere principal inertia about $\mathbf{i}_S$ (kg·m^2)
$I_{yy,S}$	0.016	Sphere principal inertia about $\mathbf{j}_S$ (kg·m^2)
$I_{zz,S}$	0.02	Sphere principal inertia about $\mathbf{k}_S$ (kg·m^2)

Model Simplifications

To reduce complexity while preserving the dominant dynamics, the following assumptions were adopted:

- The sphere CM, hull CM, and full-system CM are assumed colinear along the $\mathbf{k}_H$ axis.
- The main- and jib-sail quarter-chord points are assumed to be aligned with the sphere CM in the $\mathbf{k}_H$ direction (i.e., no vertical moment arm relative to the sphere CM).

- Aerodynamic torques acting directly on the sphere are neglected because the aerodynamic forces on the spherical structure act at or very near the sphere's CM, producing a negligible moment arm.
- α_m and α_j were treated as controlled inputs, ie no change in the angles during the course of simulation.

No-Slip Post-Stabilization

The rover operates in a rolling contact where slip at the ground contact point is not expected under nominal conditions. However, due to numerical drift and coupled multibody dynamics, small slip velocities can accumulate during time integration. To enforce the no-slip rolling condition in simulation, a post-stabilization correction is applied to the translational generalized speeds after each integration step.

Specifically, the translational generalized speeds (u, v, w) (expressed in the hull frame) are recomputed such that the ground-contact point has zero translational velocity relative to the inertial frame,

$$v_x = 0, \qquad v_y = 0, \qquad v_z = 0,$$

together with an additional algebraic relation consistent with the contact kinematics. These constraints are solved algebraically at each time step to obtain corrected values of (u, v, w) (and the associated dependent speed), and the corrected values are substituted back into the state vector before proceeding to the next step of integration. This approach acts as a post-stabilization step that suppresses constraint drift while keeping the integration scheme explicit. A similar post-stabilization strategy for enforcing kinematic constraints in multibody dynamics is discussed in Cline and Pai (2003).

First-Order State Form

The governing dynamics are implemented as two coupled sets of first-order differential equations in the generalized coordinates and generalized speeds. The generalized coordinate vector is

$$\mathbf{q} = \begin{bmatrix} q_1 & q_2 & q_3 & q_4 & q_5 & q_6 & q_7 \end{bmatrix}^T = \begin{bmatrix} x & y & z & \psi & \phi & \theta_H & \theta_S \end{bmatrix}^T, \tag{5.198}$$

with generalized velocities

$$\dot{\mathbf{q}} = \begin{bmatrix} \dot{x} & \dot{y} & \dot{z} & \dot{\psi} & \dot{\phi} & \dot{\theta}_H & \dot{\theta}_S \end{bmatrix}^T. \tag{5.199}$$

The generalized speed vector is

$$\mathbf{u} = \begin{bmatrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 & u_7 \end{bmatrix}^T = \begin{bmatrix} u & v & w & \omega_z & \omega_x & \omega_{yH} & \omega_{yS} \end{bmatrix}^T, \tag{5.200}$$

and the full state is advanced by integrating

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}(\mathbf{q}, \mathbf{u}), \qquad \dot{\mathbf{u}} = \dot{\mathbf{u}}(\mathbf{q}, \mathbf{u}). \tag{5.201}$$

The explicit expressions for $\dot{\mathbf{q}}(\mathbf{q}, \mathbf{u})$ follow from the inverse speed mapping $\dot{\mathbf{q}} = \mathbf{W}\mathbf{u}$, while $\dot{\mathbf{u}}(\mathbf{q}, \mathbf{u})$ is obtained by solving the coupled Kane equations together with the contact constraint equation.

5.6.2 Simulation Results

Following are the results:

Case 1: Wind approaching from the $-Y$ direction with initial conditions $z = 0.2$ m and all other states set to zero. The main and jib sail angles were $\alpha_m = \alpha_j = 10°$ with freestream velocity $V_\infty = 5$ m/s. The simulation shows the rover accelerating primarily in the x direction while the negative v velocity component indicates motion in the downwind direction. The Euler angles show minimal variation in ψ as it goes from 0° to about −40° , while θ_S increases steadily, indicating continuous rotation of the sphere.

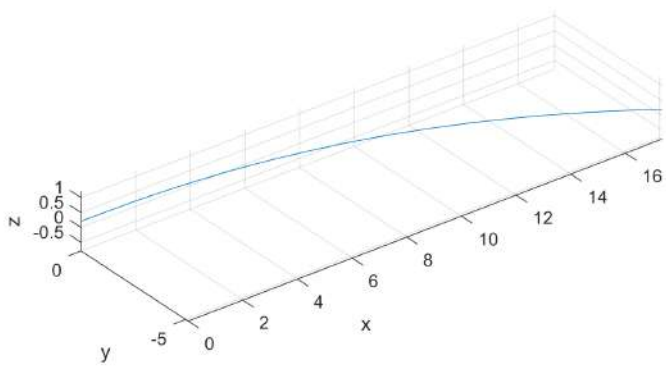
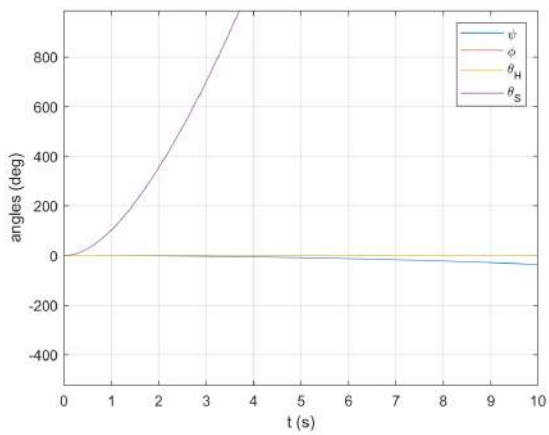
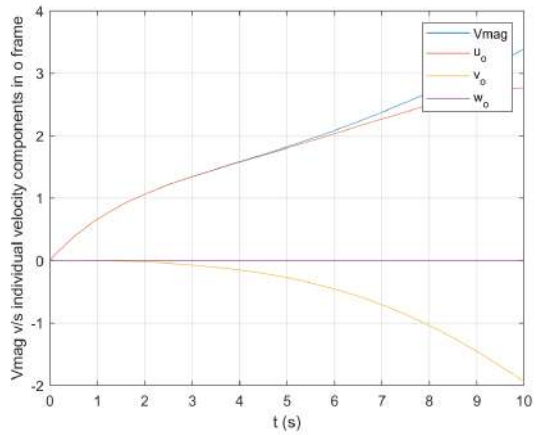


Figure 5.5: Case 1 Position



(a) case 1: Euler Angles



(b) Case 1: Velocity in IRF

Case 2: For the case with Wind approaching from the $-Y$ direction with initial conditions $z = 0.2$ m an initial yaw angle of $\psi = 20°$. The main and jib sail angles were $\alpha_m = \alpha_j = 10°$ with freestream velocity $V_\infty = 5$ m/s. The rover initially moves downwind as indicated by the negative v_O velocity component. Around $t \approx 3.5$ s, the rover briefly achieves upwind motion, which is evident from the positive v_O velocity and the corresponding change in θ_S in the Euler angle plot. However, as ψ gradually decreases and becomes negative near $t \approx 7$ s, the rover transitions back to downwind motion.

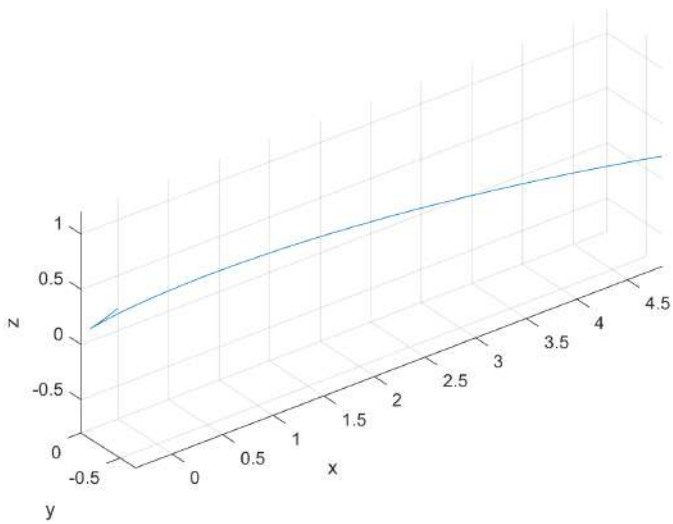
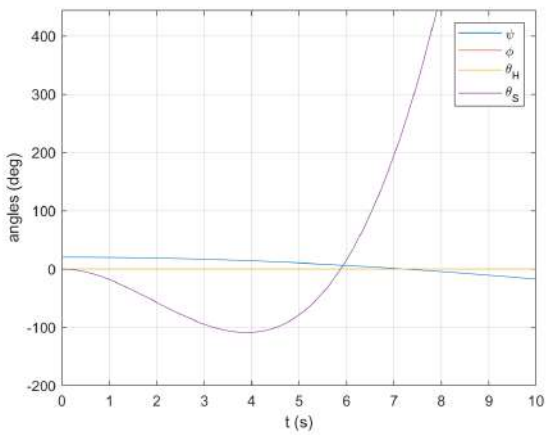
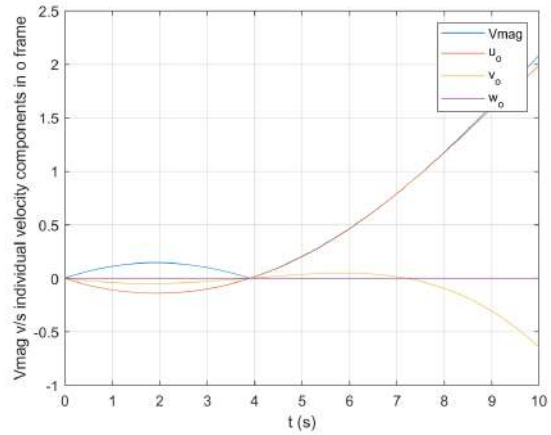


Figure 5.7: Case 2 Position



(a) case 2: Euler Angles



(b) Case 2: Velocity in IRF

Case 3: V_∞ in the $-X$ direction, with $Z = 0.2$, with initial $\psi = 40°$. The main and jib sail angles were $\alpha_m = 16°, \alpha_j = 12°$ with freestream velocity $V_\infty = 5$ m/s. This case replicates one of the wind tunnel operating conditions corresponding to the maximum lift-to-drag (L/D) ratio obtained during the aerodynamic experiments. The simulation results indicate that the rover experiences predominantly downwind motion, which can be observed from the velocity components where the rover accelerates in the wind direction. The magnitude of the velocity increases steadily over time, indicating continuous aerodynamic forcing from the sails.

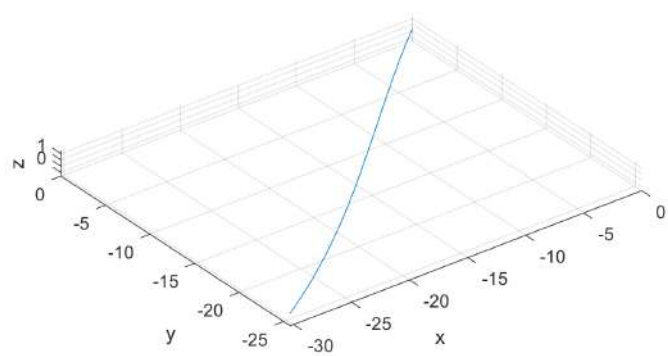
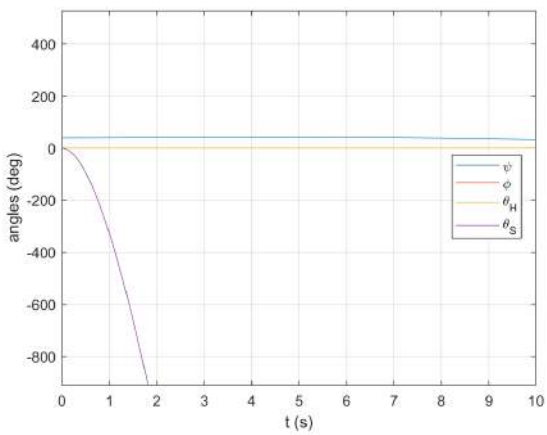
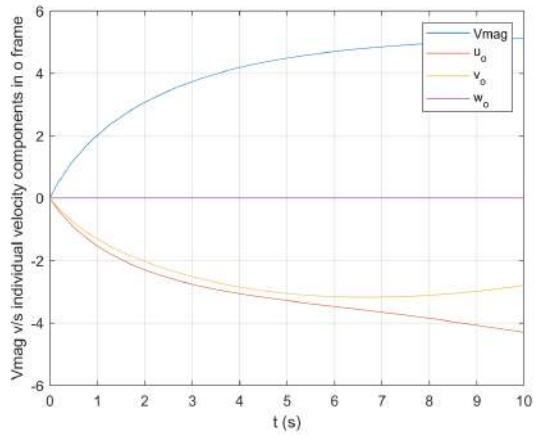


Figure 5.9: Case 3 Position



(a) case 3: Euler Angles



(b) Case 3: Velocity in IRF

Case 4: This case was performed with an incoming wind of $V_\infty = 5$ m/s in the $-Y$ direction, sail settings $\alpha_m = \alpha_j = 10°$, and initial conditions $\psi = 20°$, $\theta_H = 10°$. The velocity response indicates predominantly upwind motion over the simulated interval, as v_O remains positive while u_O stays negative and $w_O \approx 0$. The Euler-angle histories show small oscillatory variations in ϕ, and θ_H, while θ_S evolves rapidly and exhibits large excursions, consistent with sustained internal rotation of the hull relative to the sphere. This coupling between θ_S and the body kinematics drives the observed oscillatory components in the horizontal velocity response and the corresponding curved ground-track in the x–y plane.

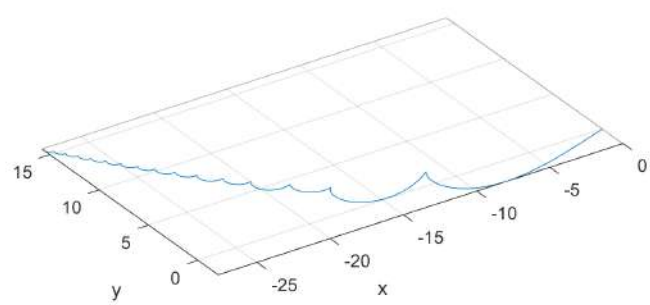
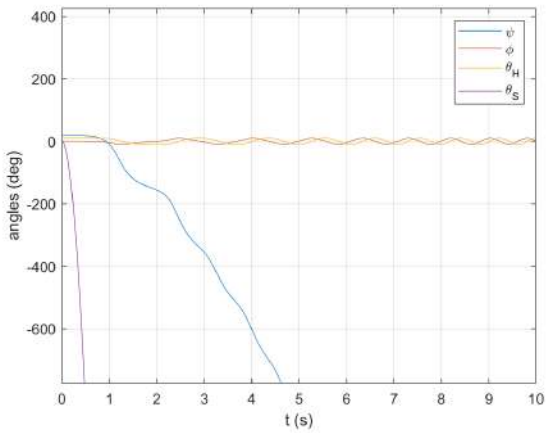
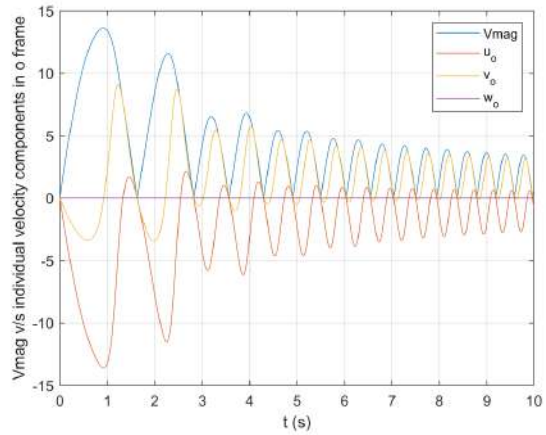


Figure 5.11: Case 4 Position



(a) case 4: Euler Angles



(b) Case 4: Velocity in IRF

CHAPTER

6

CONCLUSIONS

6.1 Summary of Results

This thesis presents the aerodynamic characterization and dynamic modeling of the SSailOR rover, a wind-propelled spherical rover designed to utilize a dual-sail configuration similar to modern sailboats. The primary objective of this work was to understand the aerodynamic performance of the rover, develop accurate aerodynamic force models, and create a dynamic simulation framework capable of predicting rover motion under varying conditions.

The study began with experimental aerodynamic characterization through wind tunnel testing. Two separate wind tunnel campaigns were conducted to evaluate the aerodynamic performance of the rover. The first test was performed on an initial baseline design of the SSailOR. The experimental data obtained from this test revealed limitations in the aerodynamic performance of the rover, particularly with respect to drag generated by structural components and limited lift generation from the sails. Based on the insights obtained from the first wind tunnel test, several design modifications were introduced to improve the aerodynamic efficiency of the rover. These modifications included increasing the sail area to enhance lift generation, reducing the hull frontal area to minimize form drag, and redesigning the traction hoop from a solid structure to a segmented three-ring configuration to reduce flow obstruction and wake interaction with the sails. A second wind tunnel test was then conducted on the improved

design to evaluate the effectiveness of these changes. The experimental results demonstrated a noticeable improvement in aerodynamic performance, with higher lift generation and reduced drag compared to the baseline configuration.

To further understand the aerodynamic behavior of the rover, computational fluid dynamics (CFD) simulations were performed using ANSYS Fluent. The simulations replicated the wind tunnel configuration, including the wind tunnel domain and rover geometry, allowing direct comparison between numerical and experimental results. The CFD analysis provided additional insight into flow structures around the rover and validated the general aerodynamic trends observed in the wind tunnel experiments.

In addition to CFD analysis, a numerical aerodynamic model for the sails was developed using the Weissinger vortex lattice method (VLM). This method was used to generate aerodynamic data across a wide range of sail angles of attack and yaw conditions. The VLM results were further validated using XFLR5 simulations to ensure the reliability of the aerodynamic predictions. These aerodynamic analyses helped establish relationships between sail angles, yaw angle, and the resulting lift and drag forces.

The aerodynamic coefficients used in the dynamic model were ultimately identified from the experimental results obtained in Wind Tunnel Test 2. These experimentally derived coefficients provided realistic aerodynamic force models for both the main and jib sails, as well as drag characteristics for non-lifting components of the rover structure.

Finally, a dynamic model of the SSailOR rover was developed using Kane's method. The rover was modeled as a coupled multibody system consisting of the spherical shell and internal hull structure. The dynamic model incorporates aerodynamic forces, gravitational forces, internal contact forces between the hull and sphere, and ground contact forces. The resulting equations of motion were implemented in a numerical simulation framework, where the rover motion can be analyzed under different wind conditions and sail configurations.

Together, the experimental testing, aerodynamic modeling, CFD analysis, and dynamic simulation provide a comprehensive framework for understanding the aerodynamic performance and motion characteristics of the SSailOR rover. These results establish the foundation for future control system development and autonomous wind-driven navigation of the rover.

6.2 Future Work

While this work provides a comprehensive aerodynamic characterization and dynamic model of the SSailOR rover, several areas remain for further development. Future efforts will focus on improving the fidelity of the aerodynamic modeling, expanding experimental validation, and developing control strategies for autonomous rover operation.

One area of future work involves refining the aerodynamic force models used in the dynamic simulation. In particular, the treatment of lateral aerodynamic forces generated by non-lifting components such as the traction hoop, ribs, and structural elements can be improved. Incorporating more detailed aerodynamic representations of these components will help capture additional force contributions that influence rover motion, especially under varying yaw angles and wind conditions.

Further improvements are also planned in the computational fluid dynamics (CFD) modeling of the rover. Higher-fidelity CFD simulations will be performed to better resolve the complex flow interactions between the sails, the spherical body, and the surrounding structural components. In addition, future simulations will investigate a wider range of sail angles of attack to more accurately characterize stall behavior and post-stall aerodynamic performance. A dynamic CFD framework is currently under development that incorporates the rotational motion of the rover cage and the interaction between moving structural components and the surrounding airflow. This approach is expected to provide deeper insight into the aerodynamic forces acting on individual rover components and help improve the accuracy of the aerodynamic coefficients used in the dynamic model.

Additional experimental validation will also be pursued through the development of a full-scale SSailOR prototype. Outdoor testing in natural wind environments will allow the evaluation of rover performance under realistic operating conditions and will provide further data to refine the aerodynamic models and simulation framework. These experiments will be particularly important for demonstrating sustained upwind mobility and validating the rover's predicted dynamic behavior.

Finally, future work will extend the current dynamic model toward the development of control strategies for autonomous navigation. The dynamic framework developed in this work provides the necessary foundation for designing guidance and control algorithms that regulate sail angles and rover heading in response to changing wind conditions. Such control strategies will enable the rover to maintain stable trajectories and achieve sustained upwind motion in practical deployment scenarios.

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APPENDICES

APPENDIX

A

ACRONYMS

A summary of all acronyms is documented in Table A.1.

Table A.1: A summary of acronyms used in alphabetical order.

Acronym	Abbreviation
Aerodynamic Influence Coefficient (matrix)	AIC
Angle of Attack	AoA
Analog Input (DAQ channel notation)	AI
Aspect Ratio	AR
Computer Aided Design	CAD
Computational Fluid Dynamics	CFD
Counter Output (DAQ channel notation)	CO
Data Acquisition	DAQ
Fused Deposition Modeling (3D printing)	FDM
Inertial Measurement Unit	IMU
Inter-Integrated Circuit (communication protocol)	I2C
Lifting Line Theory	LLT
National Instruments	NI
National Instruments PXI Express (DAQ platform)	PXIe

NACA 0020 airfoil section	NACA0020
NACA 0025 airfoil section	NACA0025
NACA 0030 airfoil section	NACA0030
Pulse Width Modulation	PWM
Pounds per square foot (dynamic pressure units)	PSF
Revolutions Per Minute	RPM
Reynolds-Averaged Navier–Stokes	RANS
Shear Stress Transport (turbulence model family)	SST
Software-in-the-Loop	SIL
Spherical Sailing Omnidirectional Rover	SSailOR
Universal Asynchronous Receiver Transmitter	UART
Universal Serial Bus	USB
Vortex Lattice Method	VLM
Wind Tunnel	WT

APPENDIX

B

VARIABLES

A summary of all variables is documented in Table B.1.

Table B.1: A summary of variables and symbols used in alphabetical order.

Variable	Symbol
Angle of attack (general)	α
Angle of attack of jib sail	α_J
Angle of attack of main sail	α_M
Apparent / freestream wind speed	V_∞
Aspect ratio	AR
Chord length	c
Dynamic pressure	Q
Drag force (global)	D
Drag coefficient (general)	C_D
Lift force (global)	L
Lift coefficient (general)	C_L
Force components measured by load cell	F_x, F_y, F_z
Moment components measured by load cell	M_x, M_y, M_z
Freestream density	ρ

Gravity	g
Rotational speed (revolutions per minute)	RPM
Time	t
Yaw angle (rover heading)	ψ
Pitch angle	ϕ
Hull rotation angle	θ_H
Sphere rotation angle	θ_S
Generalized position coordinates	$x,\ y,\ z$
Generalized translational speeds	$u,\ v,\ w$
Angular velocity components	$\omega_x,\ \omega_y s,\ \omega_z\ \omega_y h$
Circulation (Weissinger)	Γ
Downwash (Weissinger)	w